Integrated SCFA and Microbiome Analysis: Abbott Carbohydrate Obesity Project

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1. Introduction

This integrated report brings together the short-chain fatty acid (SCFA) metabolomics and 16S rRNA gene microbiome analyses from the Lurie-Abbott carbohydrate obesity project. It re-runs the key components of each analysis within a single, reproducible R Markdown workflow and adds new integrative components relating taxa, SCFA concentrations, and responder phenotypes.

The goal is to maintain the rigor and depth of the original analyses while ensuring that metadata, filtering, and modeling decisions are harmonized across datasets.

2. Project Objectives

In this section, we summarize the main scientific and analytical objectives, building on the SCFA project objectives and extending them to explicitly include microbiome composition, diversity, and taxon–SCFA relationships.

3. Methods

This integrated analysis combines methods from both the SCFA metabolomics and 16S rRNA gene microbiome analyses. Below we describe the key methodological components employed throughout this report.

3.1 Study Design and Sample Processing

The study design includes two groups (Control/Healthy and Case/Obese) exposed to three carbohydrate conditions: no added carbohydrate (control), rapid digestible carbohydrate (RDC), and slow digestible carbohydrate (SDC). Samples were collected at multiple timepoints (0H, 48H, and 72H for SCFA; 0H and 48H for microbiome). All analyses account for repeated measures within subjects through appropriate statistical modeling.

3.2 SCFA Quantification and Preprocessing

Short-chain fatty acids (butyrate, propionate, acetate, and others) were quantified using gas chromatography-mass spectrometry (GC-MS) with derivatization. Raw concentration data were preprocessed to remove quality control samples, handle missing values, and standardize units. Concentrations are reported in micromolar (μM) units. Data were log-transformed where appropriate for statistical analyses.

3.3 Microbiome Sequencing, Processing, and

Phyloseq Construction

16S rRNA gene sequencing was performed on fecal samples, and sequences were processed through standard quality control, denoising, and taxonomic assignment pipelines. The processed data were organized into a `phyloseq` object containing an OTU/ASV table, taxonomic classifications, sample metadata, and phylogenetic tree information. Relative abundances were calculated by normalizing counts to proportions within each sample. Absolute abundances were derived by integrating qPCR-determined total bacterial genome copies per sample with relative abundance data.

3.4 Statistical Modeling Strategies

3.4.1 Mixed-Effects Models for SCFA Concentrations

Linear mixed-effects models were fitted using `lmerTest::lmer()` to analyze SCFA concentrations as a function of group (Control vs Case), carbohydrate type, timepoint, and their interactions, with random intercepts for subject to account for repeated measures. Type III ANOVA tables were generated using `anova()` to test main effects and interactions, with p-values adjusted for multiple comparisons using the Benjamini-Hochberg (BH) false discovery rate (FDR) correction.

3.4.2 Alpha Diversity Analysis

Within-sample microbial diversity was assessed using three metrics: Observed richness (number of unique taxa), Shannon diversity index (accounts for both richness and evenness), and Simpson diversity index (emphasizes dominant taxa). Differences in alpha diversity between groups, carbohydrate conditions, and timepoints were tested using Wilcoxon rank-sum tests (Mann-Whitney U tests) with BH FDR correction for multiple comparisons.

3.4.3 Beta Diversity and Community Structure

Between-sample microbial community dissimilarity was quantified using Bray-Curtis distance on relative abundance data. Principal coordinates analysis (PCoA) was used to visualize community structure in reduced dimensions. Permutational multivariate analysis of variance (PERMANOVA) using `vegan::adonis2()` with 999 permutations was employed to test for significant differences in community composition between groups, carbohydrate conditions, and timepoints, with subject ID as a stratification variable to account for repeated measures.

3.4.4 Differential Abundance Analysis

Two complementary methods were used to identify taxa that differ significantly between conditions:

ANCOM-BC2: Analysis of Compositions of Microbiomes with Bias Correction 2 was applied using the `ANCOMBC` package. This method accounts for compositionality and provides log-fold change estimates with bias correction, standard errors, and FDR-adjusted q-values. Models included group, carbohydrate, timepoint, and their interactions as fixed effects, with subject ID as a random effect where appropriate.

MaAsLin2: Microbiome Multivariable Association with Linear Models 2 was applied using the `Maaslin2` package. This method uses linear models with total sum scaling (TSS) normalization and log transformation. It provides coefficient estimates, standard errors, p-values, and FDR-adjusted q-values. Models included the same fixed and random effects as ANCOM-BC2 analyses.

Both methods were applied to relative abundance data, and results were filtered to retain associations with $q < 0.1$ (FDR-adjusted p-value threshold).

3.4.5 Taxon–SCFA Association Analysis

To identify microbial taxa associated with SCFA concentrations, ANCOM-BC2 and MaAsLin2 were applied with butyrate or propionate concentrations as continuous predictors, along with carbohydrate type and their interactions. These analyses were performed on relative abundance data and included subject ID as a random effect to account for repeated measures.

3.4.6 Correlation Analyses

Spearman rank correlation coefficients were calculated to assess monotonic relationships between taxon abundances (e.g., *Fusicatenibacter*) and SCFA concentrations. Linear regression models were fitted to estimate R^2 values and test for linear associations. Correlations were examined at specific timepoints (e.g., 48H) and carbohydrate conditions (e.g., SDC only) to focus on biologically relevant contexts.

3.4.7 Responder vs Non-Responder Classification

Responder status was defined separately for butyrate and propionate based on the change in SCFA concentration from baseline (0H) to 48H (delta = 48H - 0H). For each analyte and carbohydrate condition, samples were classified as "Responder" if their delta value was greater than or equal to the median delta, and "Non-Responder" if below the median. This data-driven threshold allows for identification of individuals who show a substantial SCFA response to carbohydrate challenge relative to their peers in the same condition.

3.4.8 Responder Group Comparisons

Microbial composition differences between responder and non-responder groups were assessed using: - Alpha diversity comparisons with Wilcoxon rank-sum tests - Beta diversity comparisons using PERMANOVA - Differential abundance analysis using ANCOM-BC2 and MaAsLin2 to identify taxa that differ between responder and non-responder groups

3.5 Integration Methods

SCFA and microbiome data were integrated by matching samples based on subject ID, timepoint, and carbohydrate condition. Metadata were harmonized to ensure consistent group labels (Control/Healthy, Case/Obese) and carbohydrate labels (No Added Carb, RDC, SDC) across datasets. The integrated metadata were added to the phyloseq object's sample data component to enable joint analyses.

3.6 Visualization and Reporting

All visualizations were created using `ggplot2` with a consistent color palette and theme derived from the SCFA analysis. Forest plots display effect sizes (log-fold changes) with 95% confidence intervals for ANCOM-BC2 results. Volcano plots show effect sizes versus significance for MaAsLin2 results. Alluvial diagrams illustrate compositional changes over time. Statistical test results are displayed directly on plots where appropriate (e.g., Wilcoxon p-values on boxplots, R^2 values on scatter plots). All p-values are reported with FDR adjustment where multiple comparisons were performed.

4. Summary Data

We begin with a concise overview of the sample counts by obesity status (Control vs Case) and carbohydrate condition across both SCFA and microbiome datasets.

SCFA samples by group and carbohydrate

Group	Carbohydrate	N samples

Group	Carbohydrate	N samples
Control	No Added Carb	16
Control	RDC	16
Control	SDC	16
Case	No Added Carb	16
Case	RDC	16
Case	SDC	16

Microbiome samples by group, carbohydrate, and timepoint

Group	Carbohydrate	Timepoint	N samples
Healthy	No Added Carb	0H	12
Healthy	No Added Carb	48H	14
Healthy	RDC	0H	28
Healthy	RDC	48H	27
Healthy	SDC	0H	27
Healthy	SDC	48H	28
Obese	No Added Carb	0H	13
Obese	No Added Carb	48H	19
Obese	RDC	0H	28
Obese	RDC	48H	35
Obese	SDC	0H	28
Obese	SDC	48H	36

5. Metabolomics (SCFA)

5.1 Section Overview

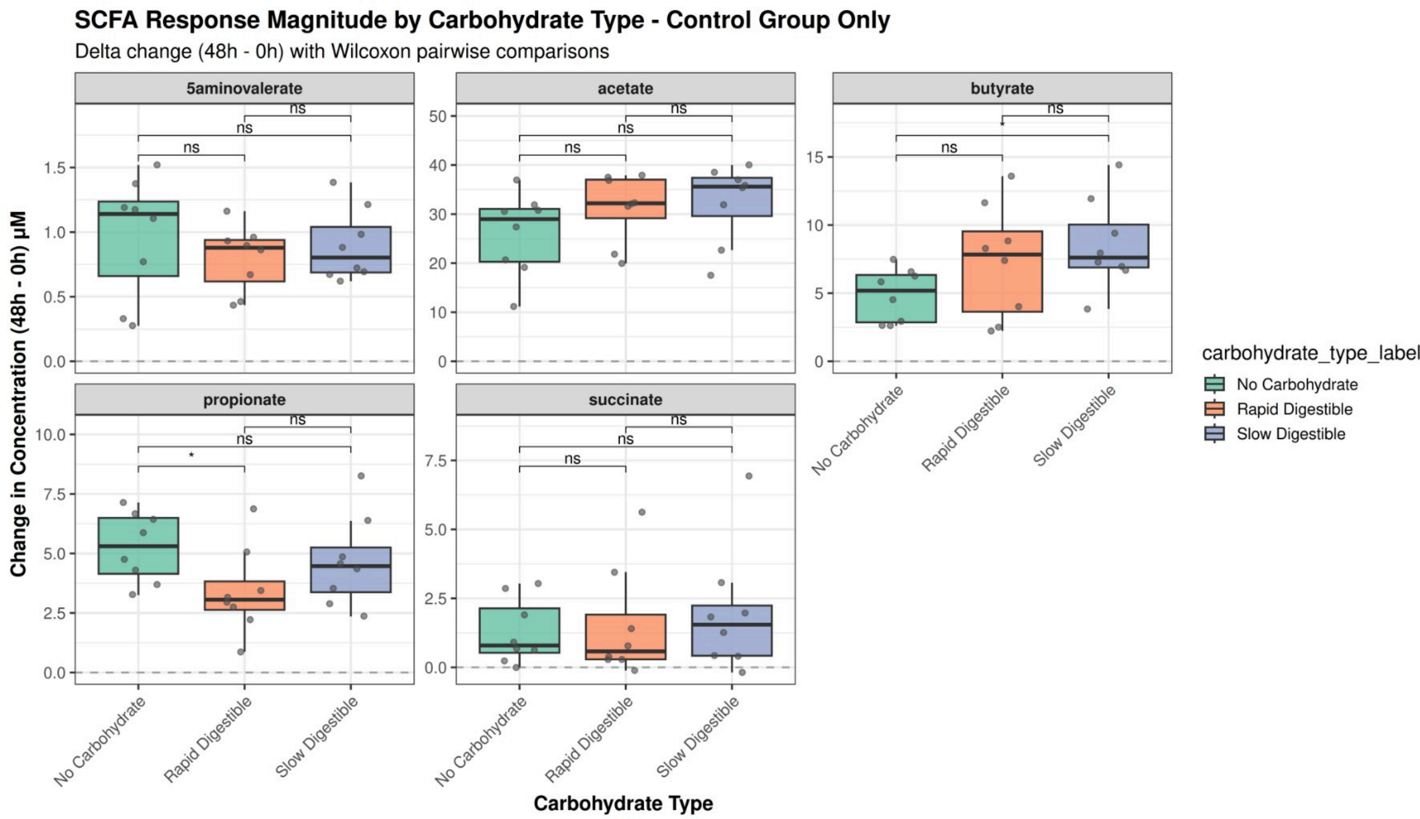
This section reproduces and extends the SCFA analyses, focusing on:

- Carbohydrate type effects (combined and stratified by group)
- SCFA concentrations over time
- Advanced mixed-effects modeling for SCFA trajectories

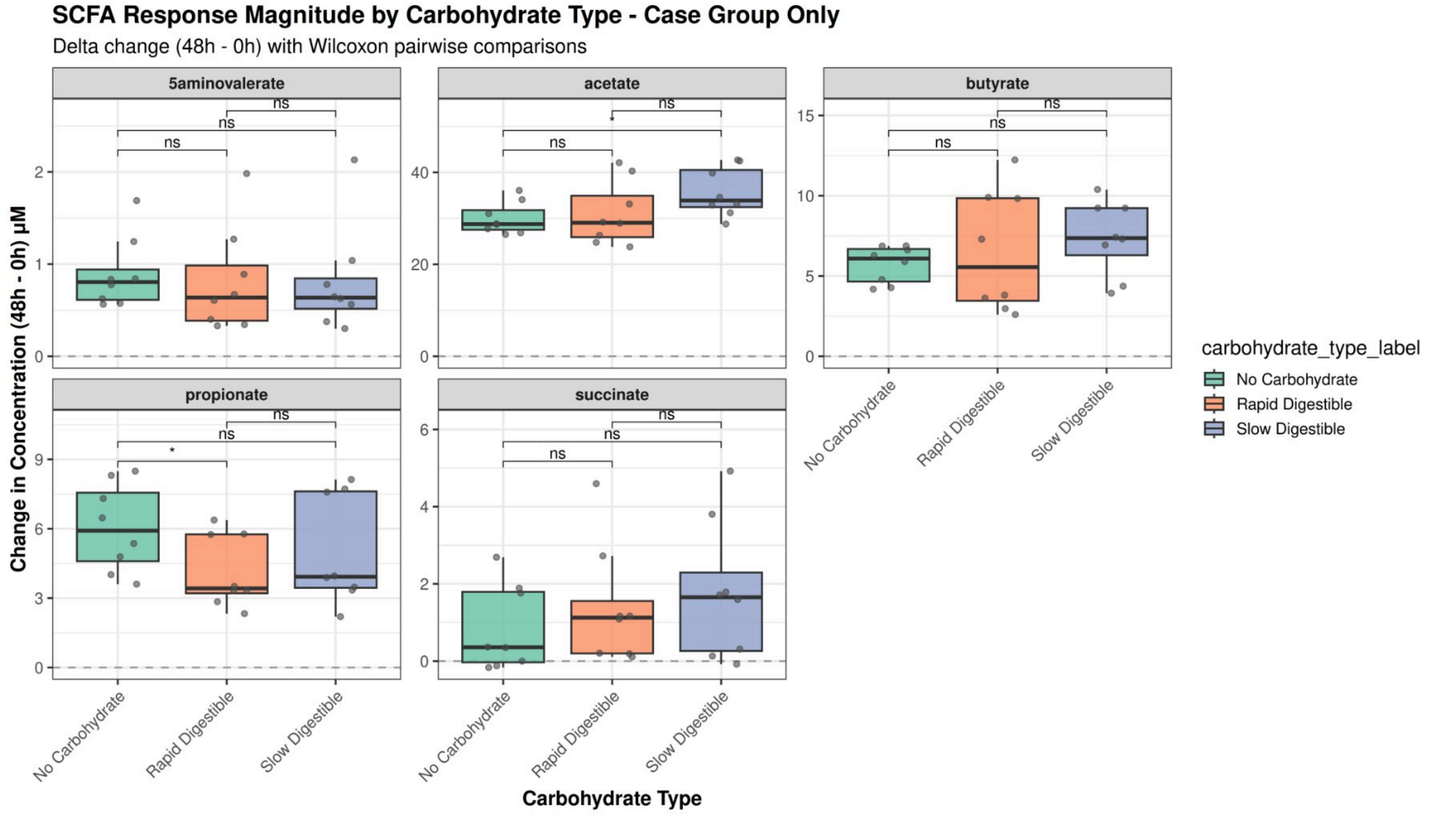
We first quantify SCFA response magnitudes (48h – 0h) and evaluate how they vary by carbohydrate type and obesity group. We then visualize SCFA concentrations over time and fit linear mixed-effects models with subject as a random effect to formally test fixed effects of group, carbohydrate type, time, and their interactions on absolute SCFA concentrations.

5.2 Carbohydrate Type Effects (Combined Groups)

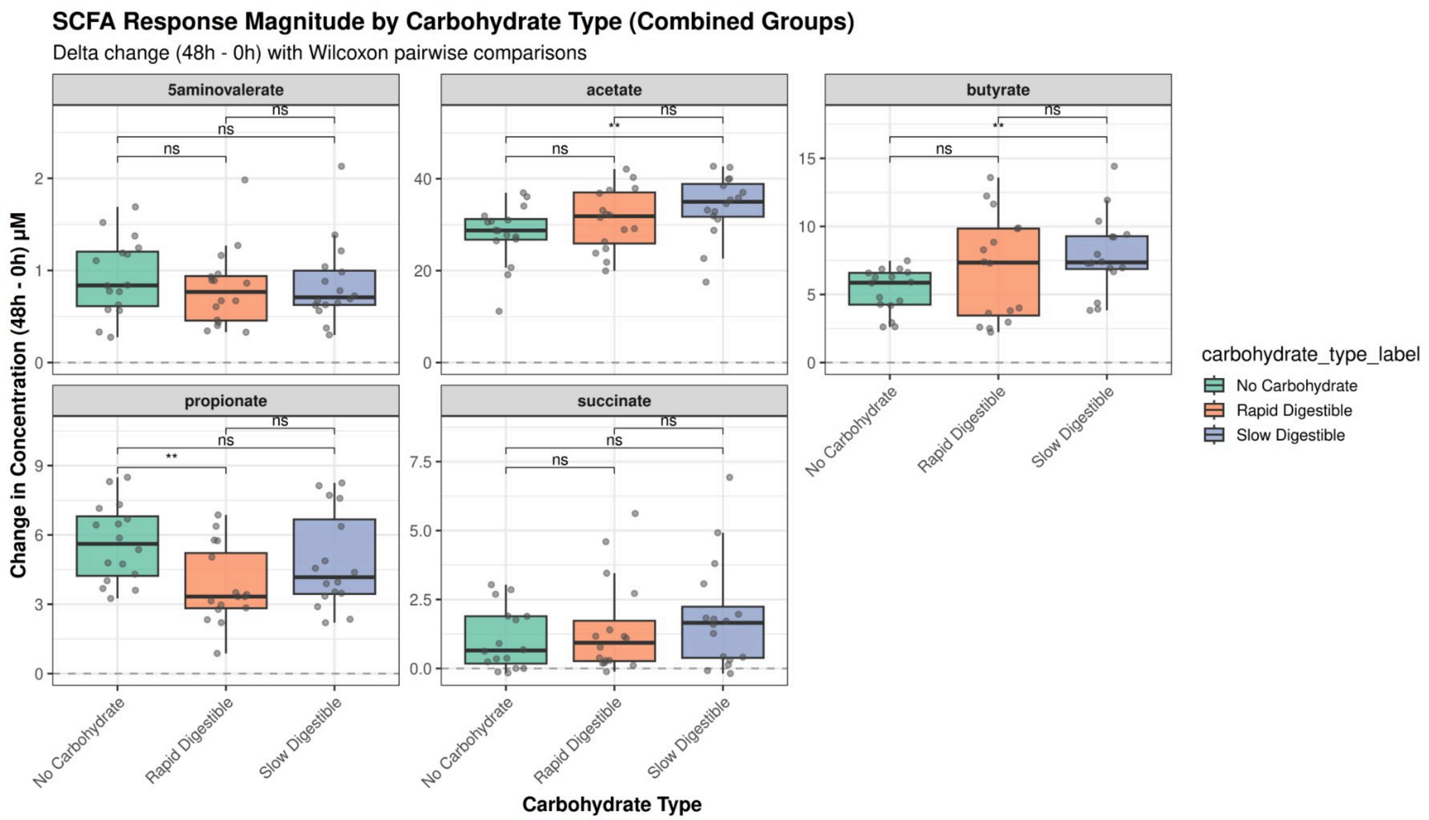
5.2.1 Control Group: Delta SCFA by Carbohydrate Type



5.2.2 Case Group: Delta SCFA by Carbohydrate Type



5.2.3 Combined Groups: Delta SCFA by Carbohydrate Type

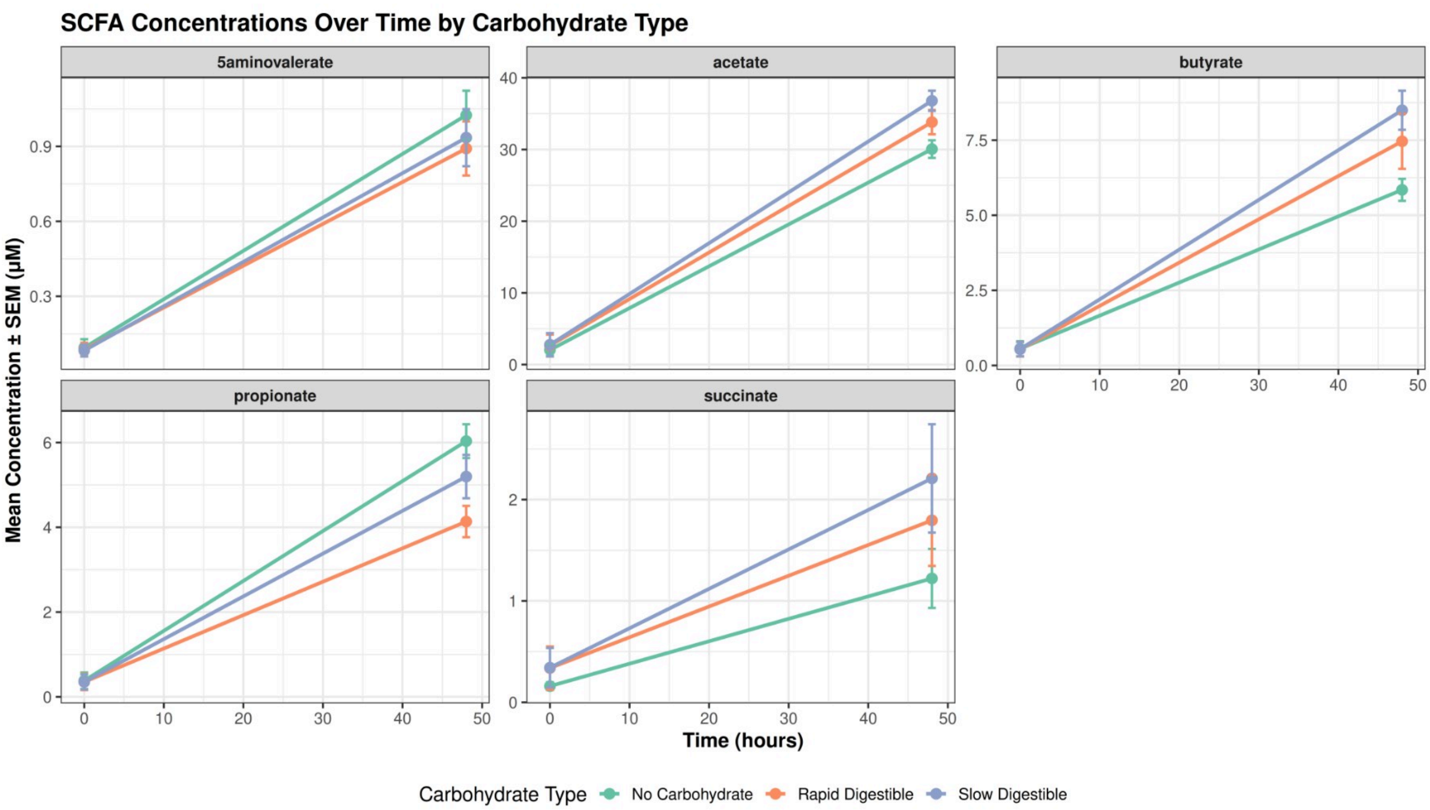


5.2.4 Statistical Summary

One-way ANOVA: Carbohydrate Type Effects on Delta Changes (Combined Groups)

analyte	Effect	p	p_adj	s
5aminovalerate	carbohydrate_type	0.7032	0.7032	ns
5aminovalerate	Residuals			ns
acetate	carbohydrate_type	0.0479	0.0479	p<0.05
acetate	Residuals			ns
butyrate	carbohydrate_type	0.0420	0.0420	p<0.05
butyrate	Residuals			ns
propionate	carbohydrate_type	0.0209	0.0209	p<0.05
propionate	Residuals			ns
succinate	carbohydrate_type	0.3920	0.3920	ns
succinate	Residuals			ns

5.3 SCFA Concentrations Over Time



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5.4 Advanced Statistical Modeling: Mixed-Effects Models

Mixed-Effects Models for SCFA Concentrations: ANOVA Table

analyte	effect	F_value	pr_f	p	
acetate	timepoint_hr	1,294.3303	0.0000	0.0000	p<0.001
propionate	timepoint_hr	472.2423	0.0000	0.0000	p<0.001
butyrate	timepoint_hr	326.8937	0.0000	0.0000	p<0.001
5aminovalerate	timepoint_hr	259.2812	0.0000	0.0000	p<0.001
succinate	timepoint_hr	46.8123	0.0000	0.0000	p<0.001
propionate	carbohydrate_type	6.4607	0.0027	0.0084	p<0.01
propionate	carbohydrate_type:timepoint_hr	6.1058	0.0036	0.0084	p<0.01
acetate	carbohydrate_type	6.2720	0.0031	0.0109	p<0.05
butyrate	carbohydrate_type	4.2545	0.0180	0.0421	p<0.05
butyrate	carbohydrate_type:timepoint_hr	4.3484	0.0166	0.0421	p<0.05
acetate	carbohydrate_type:timepoint_hr	4.0477	0.0217	0.0506	ns
propionate	group:timepoint_hr	2.1160	0.1502	0.2629	ns
succinate	carbohydrate_type	2.5593	0.0846	0.2960	ns
acetate	group:timepoint_hr	1.8526	0.1778	0.3112	ns
acetate	group:carbohydrate_type	1.5021	0.2298	0.3217	ns
butyrate	group	1.2338	0.2854	0.4994	ns
acetate	group	0.4661	0.5059	0.5767	ns
acetate	group:carbohydrate_type:timepoint_hr	0.5547	0.5767	0.5767	ns
butyrate	group:carbohydrate_type	0.7097	0.4953	0.5778	ns
butyrate	group:carbohydrate_type:timepoint_hr	0.7224	0.4892	0.5778	ns
succinate	group	0.9593	0.3440	0.6020	ns
succinate	carbohydrate_type:timepoint_hr	1.1820	0.3127	0.6020	ns
butyrate	group:timepoint_hr	0.2528	0.6167	0.6167	ns
succinate	group:timepoint_hr	0.3234	0.5714	0.7999	ns

5aminovalerate	group	0.4080	0.5333	0.9028	ns
5aminovalerate	carbohydrate_type	0.6127	0.5448	0.9028	ns
5aminovalerate	group:timepoint_hr	0.2143	0.6448	0.9028	ns
5aminovalerate	carbohydrate_type:timepoint_hr	0.4729	0.6252	0.9028	ns
5aminovalerate	group:carbohydrate_type	0.1012	0.9039	0.9146	ns
5aminovalerate	group:carbohydrate_type:timepoint_hr	0.0894	0.9146	0.9146	ns
succinate	group:carbohydrate_type	0.0680	0.9343	0.9485	ns
succinate	group:carbohydrate_type:timepoint_hr	0.0530	0.9485	0.9485	ns
propionate	group	0.0013	0.9718	0.9718	ns
propionate	group:carbohydrate_type	0.0603	0.9415	0.9718	ns
propionate	group:carbohydrate_type:timepoint_hr	0.0822	0.9211	0.9718	ns

5.5 Summary of SCFA Results

The SCFA metabolomics analysis reveals several key findings regarding short-chain fatty acid production in response to carbohydrate challenge in adolescents with and without obesity:

1. Dominant Effect of Carbohydrate Type and Time

The primary drivers of SCFA production in this ex vivo model were the type of carbohydrate supplied and the incubation time. Both rapid digestible carbohydrate (RDC) and slow digestible carbohydrate (SDC) led to significant increases in all measured SCFAs (butyrate, propionate, acetate) over 48 hours compared to the no-carbohydrate control. This was confirmed by: - Significant differences in delta changes (48H - 0H) between carbohydrate conditions across multiple SCFAs (Section 5.2) - Clear temporal increases in SCFA concentrations over time, with distinct trajectories for each carbohydrate type (Section 5.3) - Robust statistical significance in the mixed-effects models, where carbohydrate type and timepoint consistently emerged as significant main effects and interactions (Section 5.4)

2. Subtle Differences Between Case and Control Groups

While there were some minor differences in baseline SCFA levels and response patterns between the case (obesity) and control groups, the overall response to carbohydrate interventions was remarkably similar. The mixed-effects models showed limited significant main effects of `group` or major interactions involving `group` , suggesting that the fundamental capacity of the gut microbiota to ferment these carbohydrates is not dramatically altered in the context of obesity in this cohort. This finding suggests that obesity status alone does not substantially impair the microbial fermentation capacity for these carbohydrate substrates.

3. Carbohydrate-Specific Response Patterns

The analyses revealed distinct response patterns for different carbohydrate types: - Both RDC and SDC stimulated SCFA production compared to no-carbohydrate controls - The magnitude and timing of responses varied by carbohydrate type, as evidenced by significant carbohydrate × timepoint interactions in the mixed-effects models - These differences highlight the importance of carbohydrate structure and digestibility in shaping microbial metabolic output

4. Pronounced Interpersonal Variation

A central finding is the high degree of subject-to-subject variability in SCFA production. The delta change analyses (Section 5.2) and temporal trajectories (Section 5.3) clearly demonstrate that: - Some individuals are consistently high or low SCFA producers across conditions - Response patterns to different carbohydrates are unique to each individual - This variability is evident even when accounting for group (case vs control) and carbohydrate type effects

This pronounced interpersonal variation underscores the need for personalized approaches in nutrition, as a “one-size-fits-all” dietary intervention is unlikely to be effective. The identification of responder and non-responder groups (explored in Section 7) builds upon this finding to enable more targeted analyses of microbial composition differences.

5. Statistical Model Insights

The linear mixed-effects models (Section 5.4) provide formal statistical confirmation of these patterns, accounting for repeated measures within subjects. Key model findings include: - Significant main effects of carbohydrate type and timepoint across multiple SCFAs - Significant interactions between carbohydrate type and timepoint, indicating that the temporal response depends on the carbohydrate substrate - Limited group effects, consistent with the observation that obesity status does not dramatically alter fermentation capacity - The models account for subject-level variation through random intercepts, capturing the interpersonal differences observed in the data

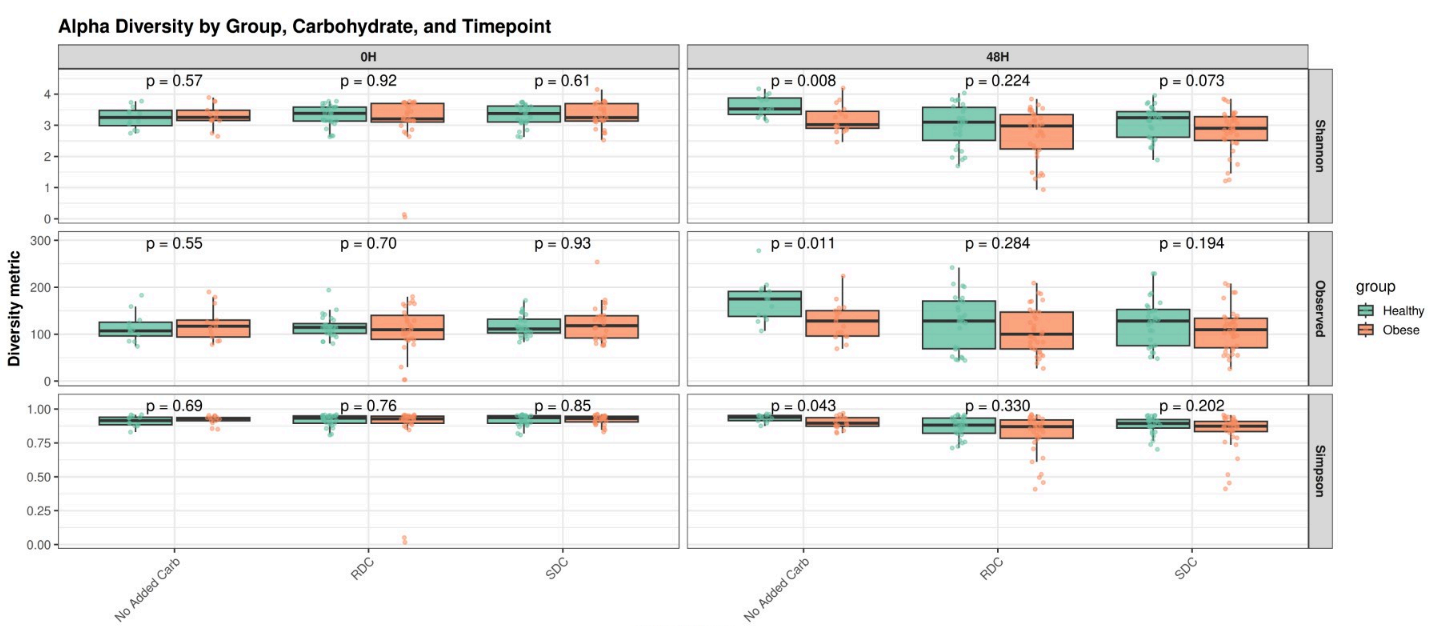
These findings set the stage for the integrated analyses in Section 7, where we examine how microbial composition relates to SCFA production patterns and identify specific taxa associated with SCFA concentrations and responder phenotypes.

6. Microbiome

6.1 Section Overview

This section reproduces key microbiome results, including alpha and beta diversity, community composition, and differential abundance, using the harmonized metadata and styling. We focus first on within-sample diversity (alpha diversity) to summarize richness and evenness patterns across groups, carbohydrate conditions, and timepoints, then move to between-sample diversity (beta diversity) and taxon-level changes.

6.2 Alpha Diversity



Alpha diversity at 48H: summary by metric, group, and carbohydrate

metric	group	carbohydrate	mean		
Shannon	Healthy	No Added Carb	3.626	0.337	13
Shannon	Healthy	RDC	2.989	0.685	27
Shannon	Healthy	SDC	3.099	0.528	28
Shannon	Obese	No Added Carb	3.191	0.444	17
Shannon	Obese	RDC	2.712	0.793	35
Shannon	Obese	SDC	2.815	0.665	36
Observed	Healthy	No Added Carb	170.462	44.612	13
Observed	Healthy	RDC	125.333	58.541	27
Observed	Healthy	SDC	124.321	50.030	28
Observed	Obese	No Added Carb	127.588	39.391	17
Observed	Obese	RDC	107.543	47.928	35
Observed	Obese	SDC	110.028	45.640	36
Simpson	Healthy	No Added Carb	0.932	0.027	13
Simpson	Healthy	RDC	0.867	0.074	27
Simpson	Healthy	SDC	0.879	0.066	28
Simpson	Obese	No Added Carb	0.899	0.045	17
Simpson	Obese	RDC	0.815	0.151	35
Simpson	Obese	SDC	0.835	0.132	36

Wilcoxon tests: Healthy vs Obese within carbohydrate × timepoint for each alpha metric

metric	carbohydrate	timepoint	comparison	p value	p adj	
Shannon	No Added Carb	0H	Healthy vs Obese	0.5740	0.9000	ns
Shannon	No Added Carb	48H	Healthy vs Obese	0.0080	0.1017	ns
Shannon	RDC	0H	Healthy vs Obese	0.9160	0.9330	ns
Shannon	RDC	48H	Healthy vs Obese	0.2240	0.5760	ns
Shannon	SDC	0H	Healthy vs Obese	0.6100	0.9000	ns
Shannon	SDC	48H	Healthy vs Obese	0.0728	0.3276	ns
Observed	No Added Carb	0H	Healthy vs Obese	0.5490	0.9000	ns

Observed	No Added Carb	48H	Healthy vs Obese	0.0113	0.1017	ns
Observed	RDC	0H	Healthy vs Obese	0.7000	0.9000	ns
Observed	RDC	48H	Healthy vs Obese	0.2840	0.6390	ns
Observed	SDC	0H	Healthy vs Obese	0.9330	0.9330	ns
Observed	SDC	48H	Healthy vs Obese	0.1940	0.5760	ns
Simpson	No Added Carb	0H	Healthy vs Obese	0.6890	0.9000	ns
Simpson	No Added Carb	48H	Healthy vs Obese	0.0433	0.2598	ns
Simpson	RDC	0H	Healthy vs Obese	0.7640	0.9168	ns
Simpson	RDC	48H	Healthy vs Obese	0.3300	0.6600	ns
Simpson	SDC	0H	Healthy vs Obese	0.8480	0.9330	ns
Simpson	SDC	48H	Healthy vs Obese	0.2020	0.5760	ns

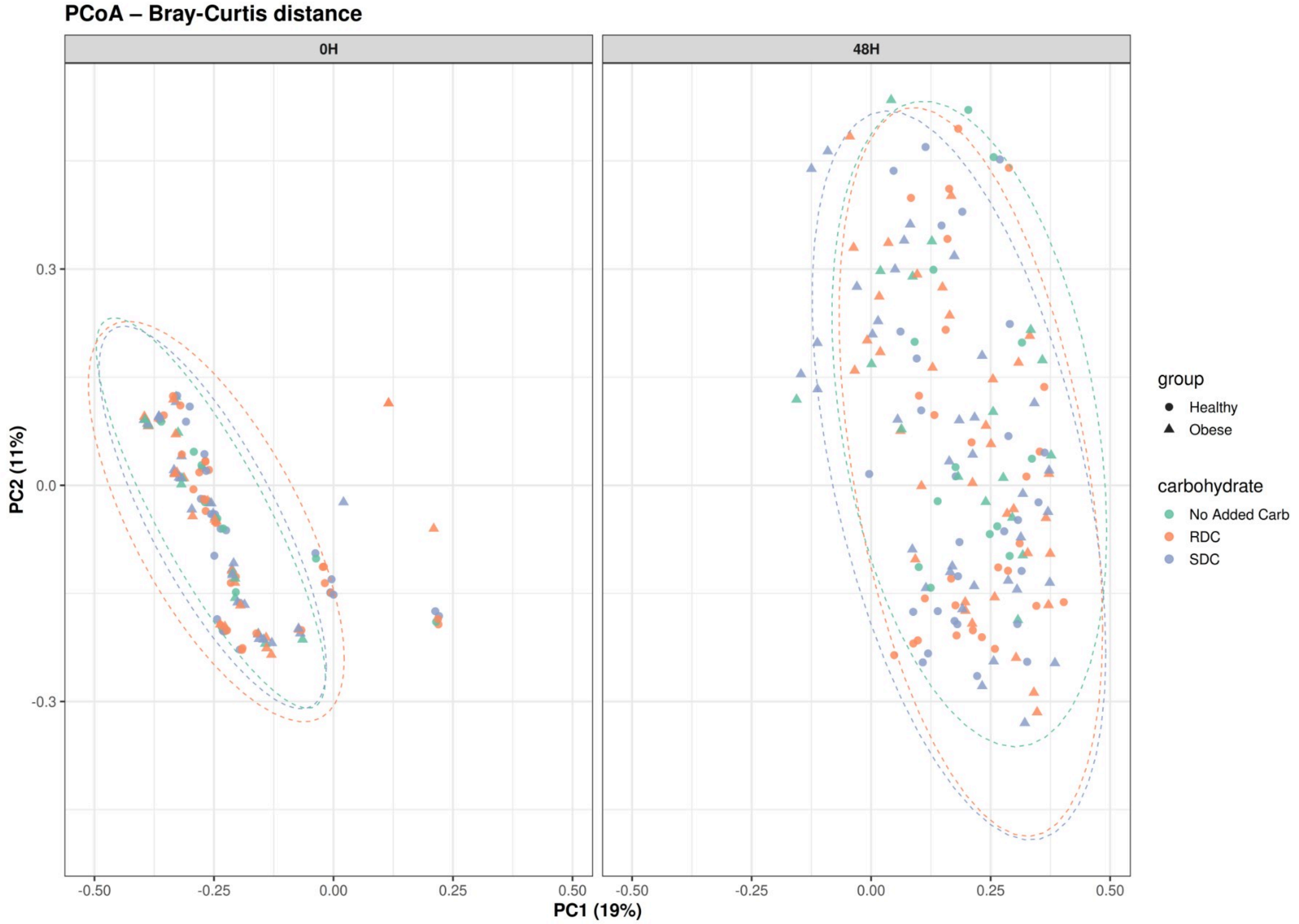
```
##
##
## #### Wilcoxon tests: Carbohydrate comparisons within group × timepoint
```

Wilcoxon tests: Carbohydrate comparisons within group × timepoint for each alpha metric

metric	group	timepoint	comparison	p value	p adju	
Shannon	Healthy	0H	No Added Carb vs RDC	0.5500	0.8250	ns
Shannon	Healthy	0H	No Added Carb vs SDC	0.5380	0.8250	ns
Shannon	Healthy	0H	RDC vs SDC	0.9670	0.9670	ns
Shannon	Healthy	48H	No Added Carb vs RDC	0.0020	0.0060	**
Shannon	Healthy	48H	No Added Carb vs SDC	0.0040	0.0060	**
Shannon	Healthy	48H	RDC vs SDC	0.5990	0.5990	ns
Shannon	Obese	0H	No Added Carb vs RDC	0.6280	0.9670	ns
Shannon	Obese	0H	No Added Carb vs SDC	0.9670	0.9670	ns
Shannon	Obese	0H	RDC vs SDC	0.6780	0.9670	ns
Shannon	Obese	48H	No Added Carb vs RDC	0.0980	0.1460	ns
Shannon	Obese	48H	No Added Carb vs SDC	0.0750	0.1460	ns
Shannon	Obese	48H	RDC vs SDC	0.7880	0.7880	ns

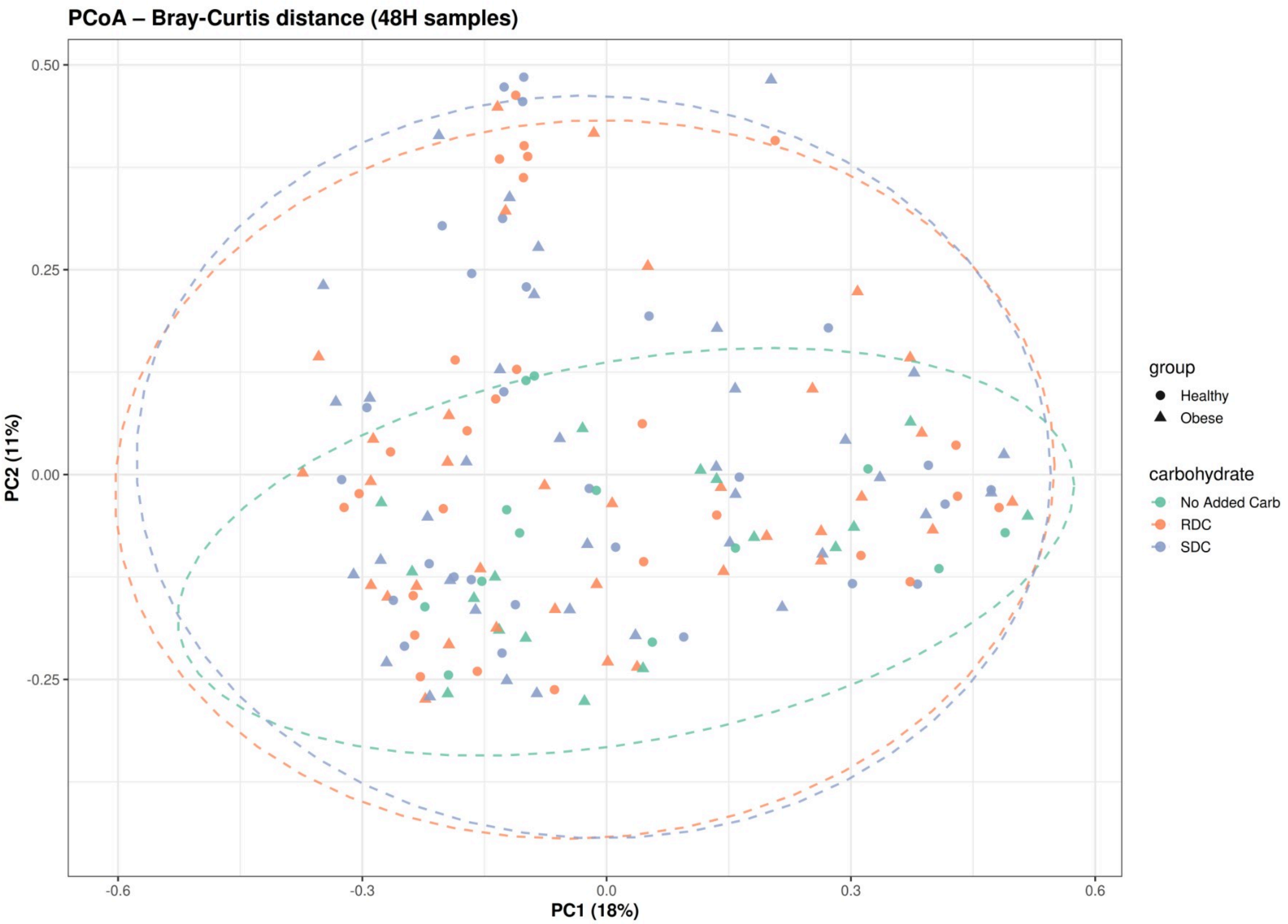
Observed	Healthy	0H	No Added Carb vs RDC	0.5450	0.8180	ns
Observed	Healthy	0H	No Added Carb vs SDC	0.5130	0.8180	ns
Observed	Healthy	0H	RDC vs SDC	0.9260	0.9260	ns
Observed	Healthy	48H	No Added Carb vs RDC	0.0330	0.0490	*
Observed	Healthy	48H	No Added Carb vs SDC	0.0080	0.0240	*
Observed	Healthy	48H	RDC vs SDC	1.0000	1.0000	ns
Observed	Obese	0H	No Added Carb vs RDC	0.5850	0.8770	ns
Observed	Obese	0H	No Added Carb vs SDC	1.0000	1.0000	ns
Observed	Obese	0H	RDC vs SDC	0.4760	0.8770	ns
Observed	Obese	48H	No Added Carb vs RDC	0.1260	0.2370	ns
Observed	Obese	48H	No Added Carb vs SDC	0.1580	0.2370	ns
Observed	Obese	48H	RDC vs SDC	0.7560	0.7560	ns
Simpson	Healthy	0H	No Added Carb vs RDC	0.5700	0.8550	ns
Simpson	Healthy	0H	No Added Carb vs SDC	0.4990	0.8550	ns
Simpson	Healthy	0H	RDC vs SDC	0.9670	0.9670	ns
Simpson	Healthy	48H	No Added Carb vs RDC	0.0020	0.0060	**
Simpson	Healthy	48H	No Added Carb vs SDC	0.0050	0.0070	**
Simpson	Healthy	48H	RDC vs SDC	0.6340	0.6340	ns
Simpson	Obese	0H	No Added Carb vs RDC	0.9890	0.9890	ns
Simpson	Obese	0H	No Added Carb vs SDC	0.8140	0.9890	ns
Simpson	Obese	0H	RDC vs SDC	0.7270	0.9890	ns
Simpson	Obese	48H	No Added Carb vs RDC	0.0660	0.1120	ns
Simpson	Obese	48H	No Added Carb vs SDC	0.0750	0.1120	ns
Simpson	Obese	48H	RDC vs SDC	0.7620	0.7620	ns

6.3 Beta Diversity



PERMANOVA results (Bray-Curtis; all timepoints, subject-level strata)

term	f	r2	
Model	9.829	0.171	0.001
Residual		0.829	
Total		1.000	

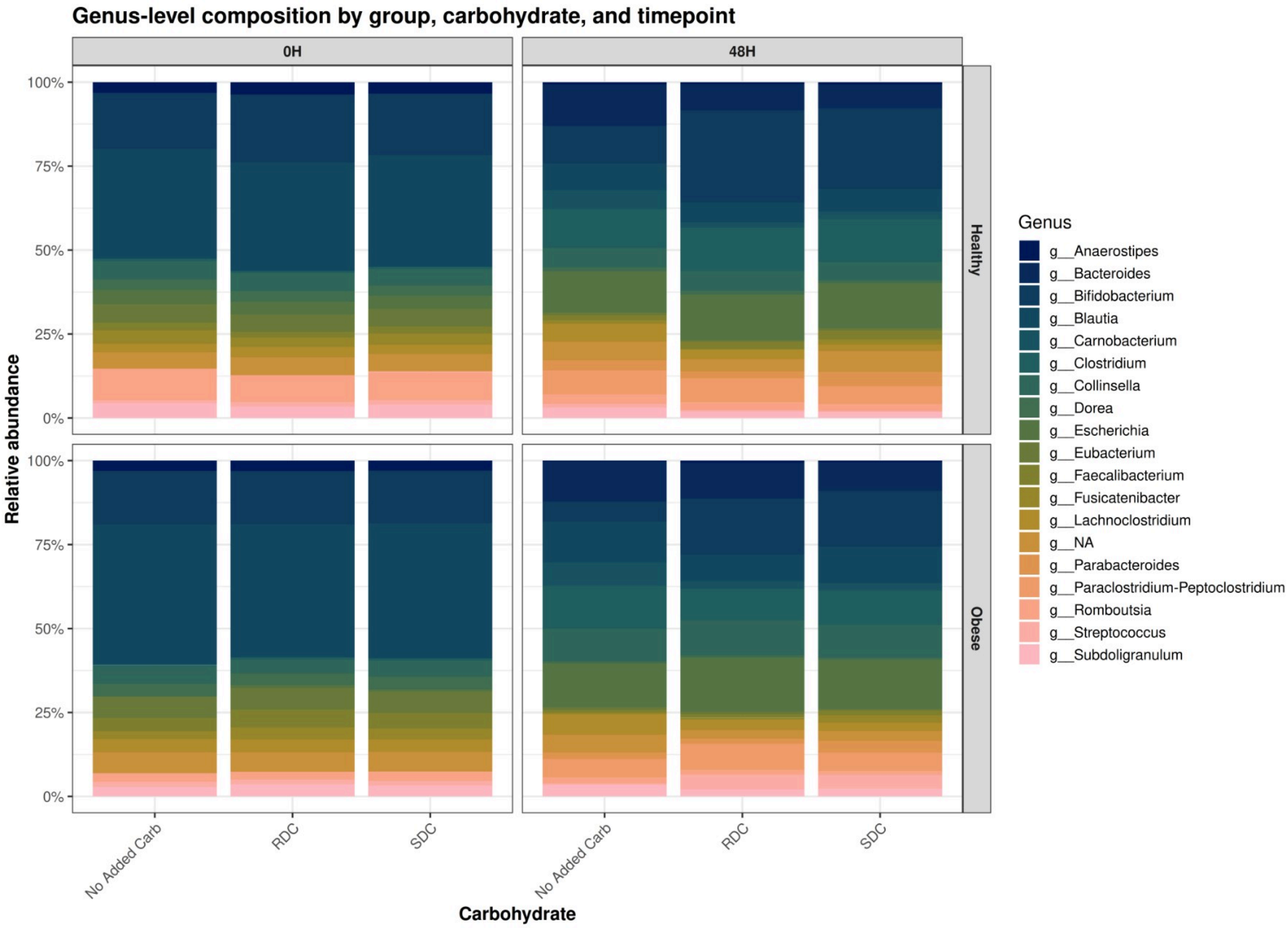


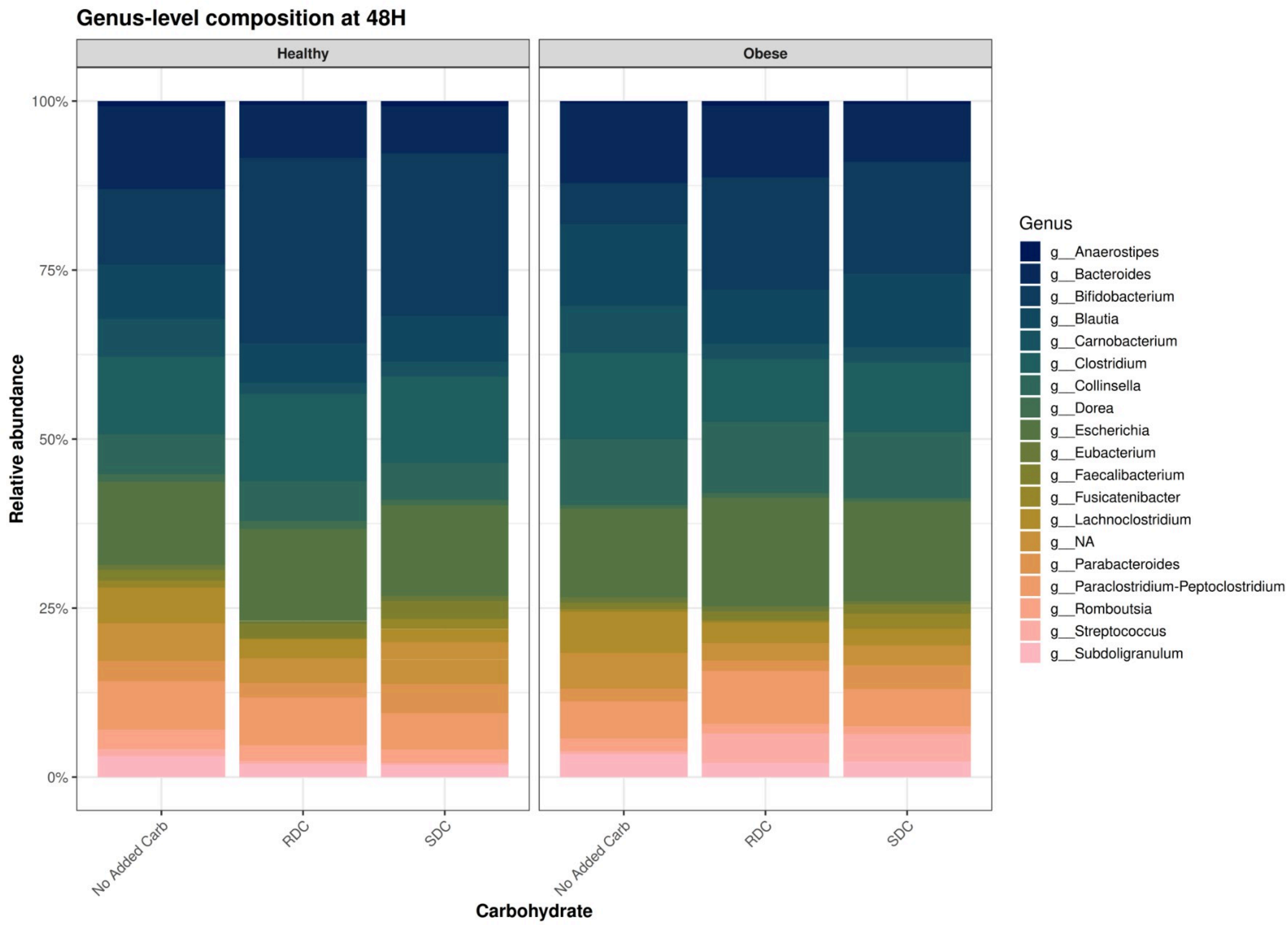
PERMANOVA results (Bray-Curtis; 48H only, subject-level strata)

term	f	r2	
Model	1.470	0.047	0.001
Residual		0.953	
Total		1.000	

6.4 Community Composition

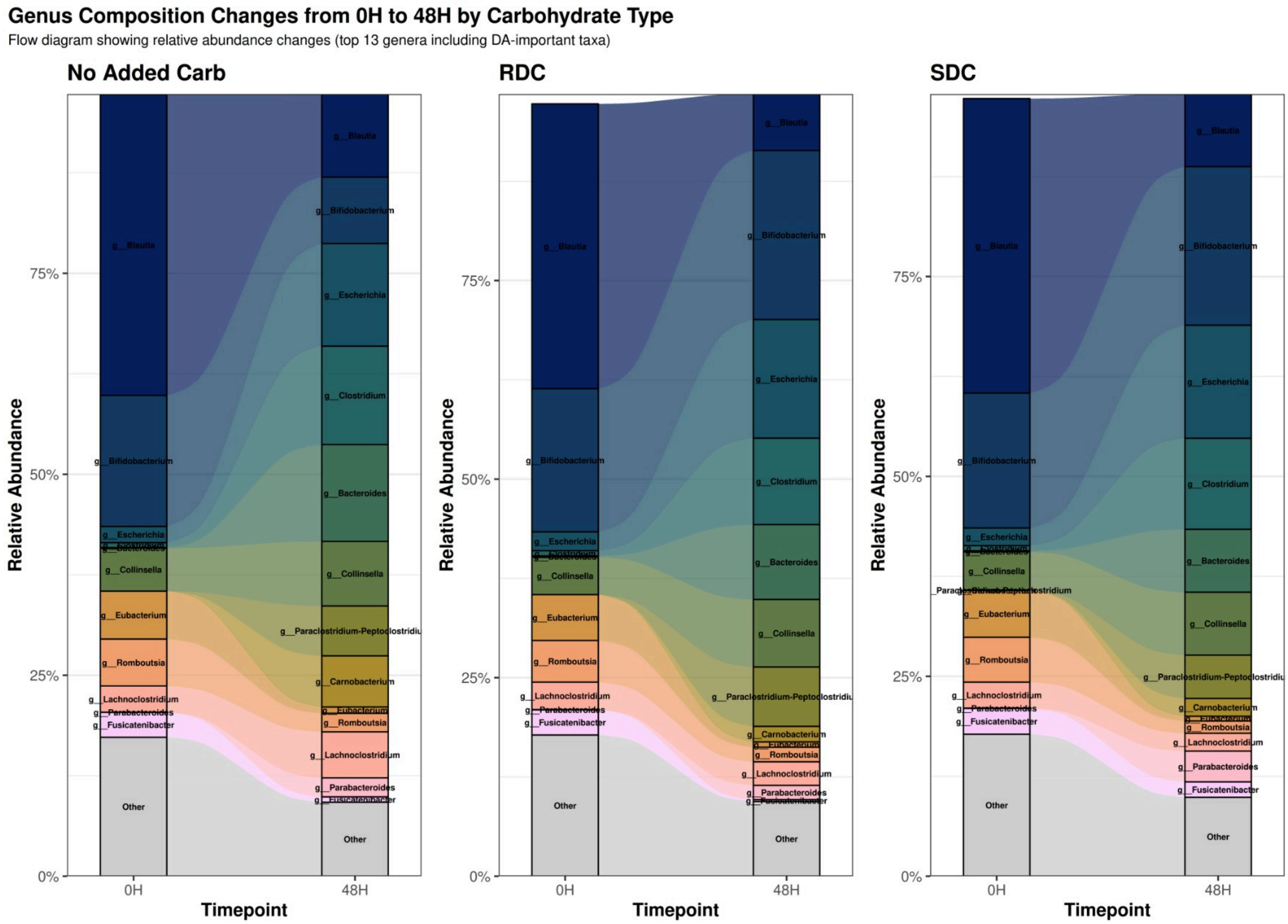
6.4.1 Genus-Level Aggregation and Overview





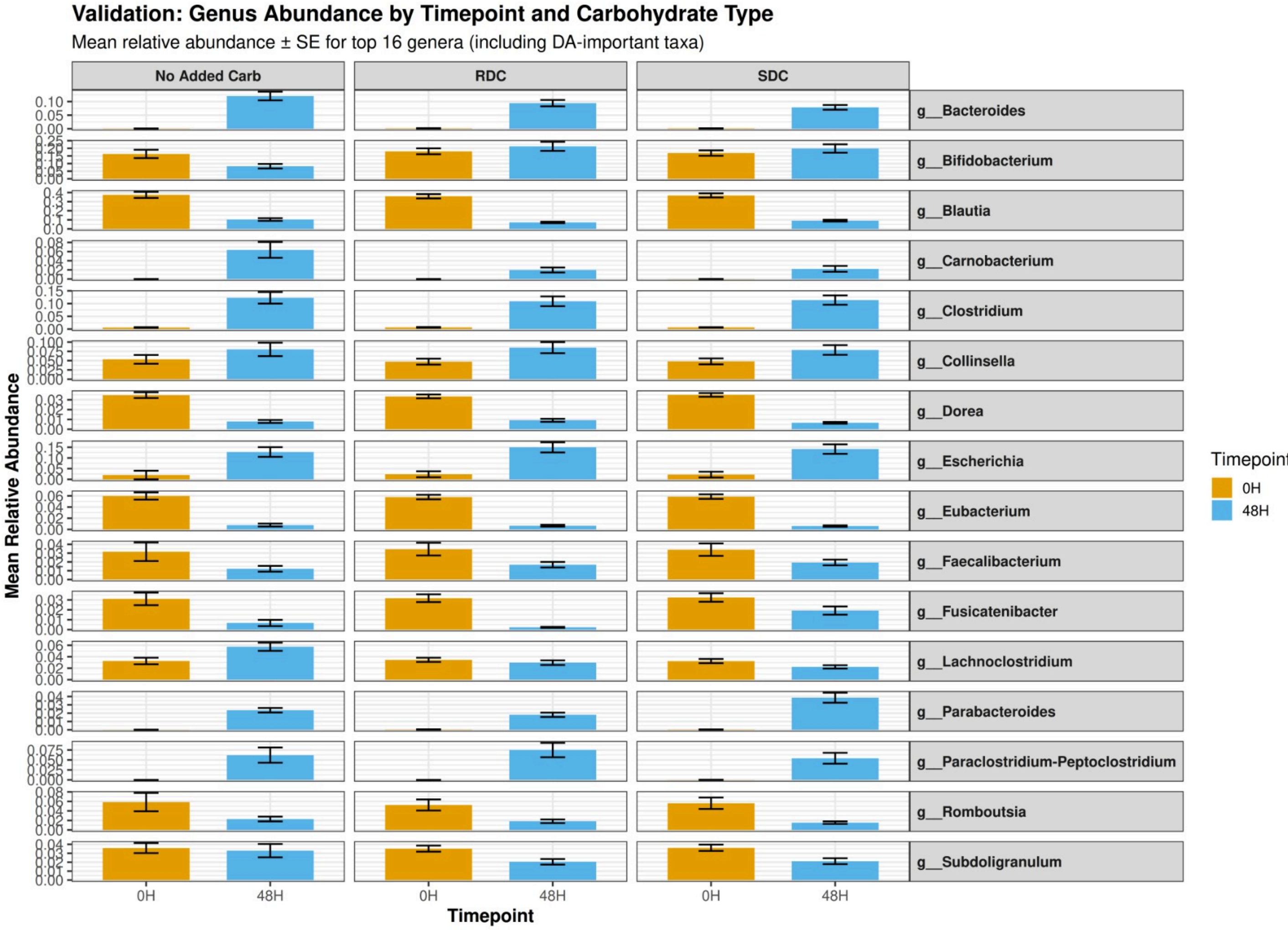
6.4.2 Genus Composition Changes from 0H to 48H by

Carbohydrate Type



6.4.3 Validation: Bar Plots of Genus Abundances by Timepoint

and Carbohydrate



6.5 Differential Abundance and Predictive Modeling

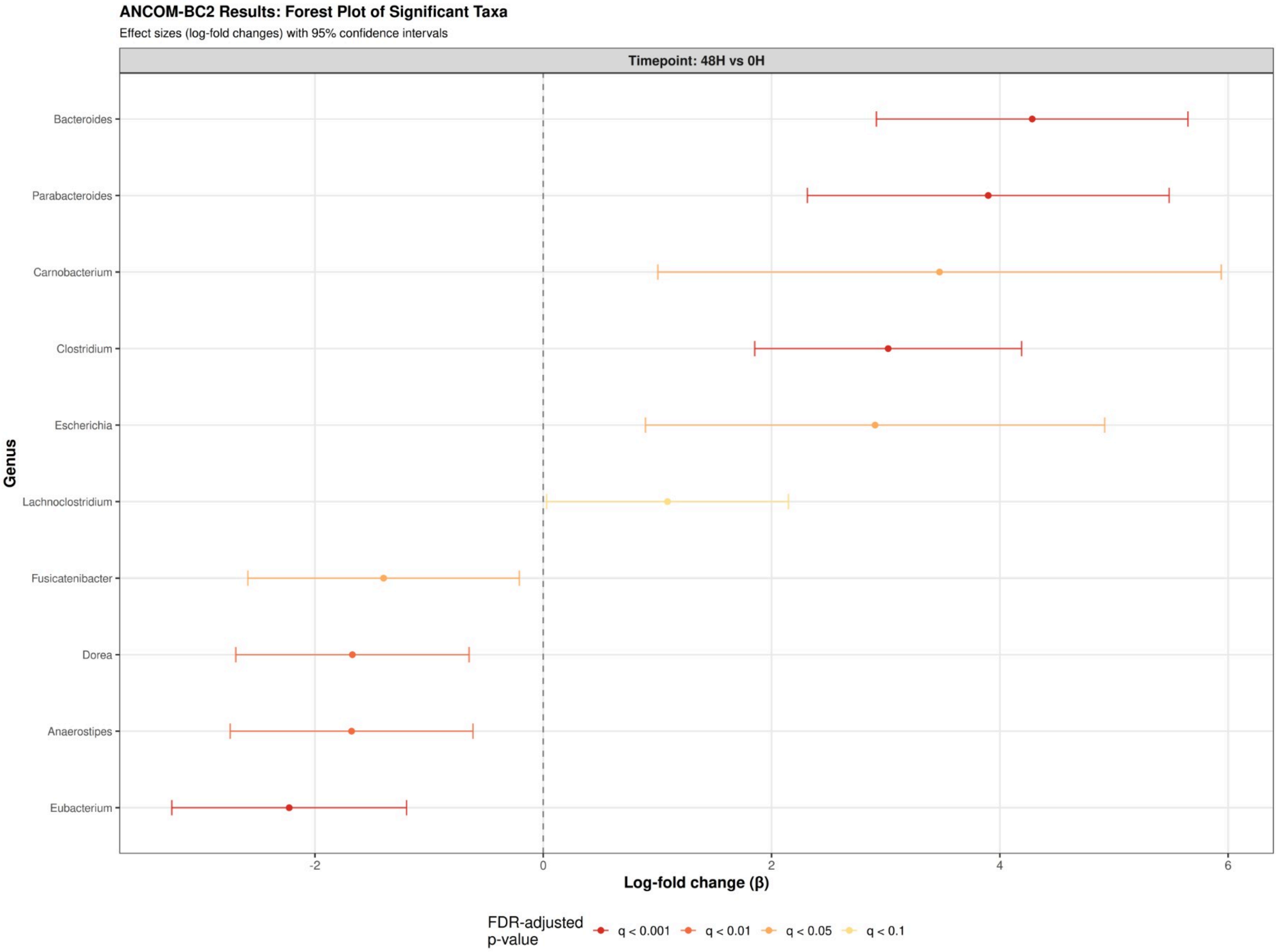
Differential abundance analysis identifies microbial taxa that differ significantly between experimental conditions. We applied multiple complementary methods to ensure robust identification of differentially abundant taxa: ANCOM-BC2 (Analysis of Compositions of Microbiomes with Bias Correction), which accounts for compositional bias and uses mixed-effects models; MaAsLin2 (Microbiome Multivariable Associations with Linear Models), which handles multivariate associations with normalization and transformations; and LEfSe (Linear Discriminant Analysis Effect Size), which identifies biomarker taxa between groups.

6.5.1 ANCOM-BC2 Analysis

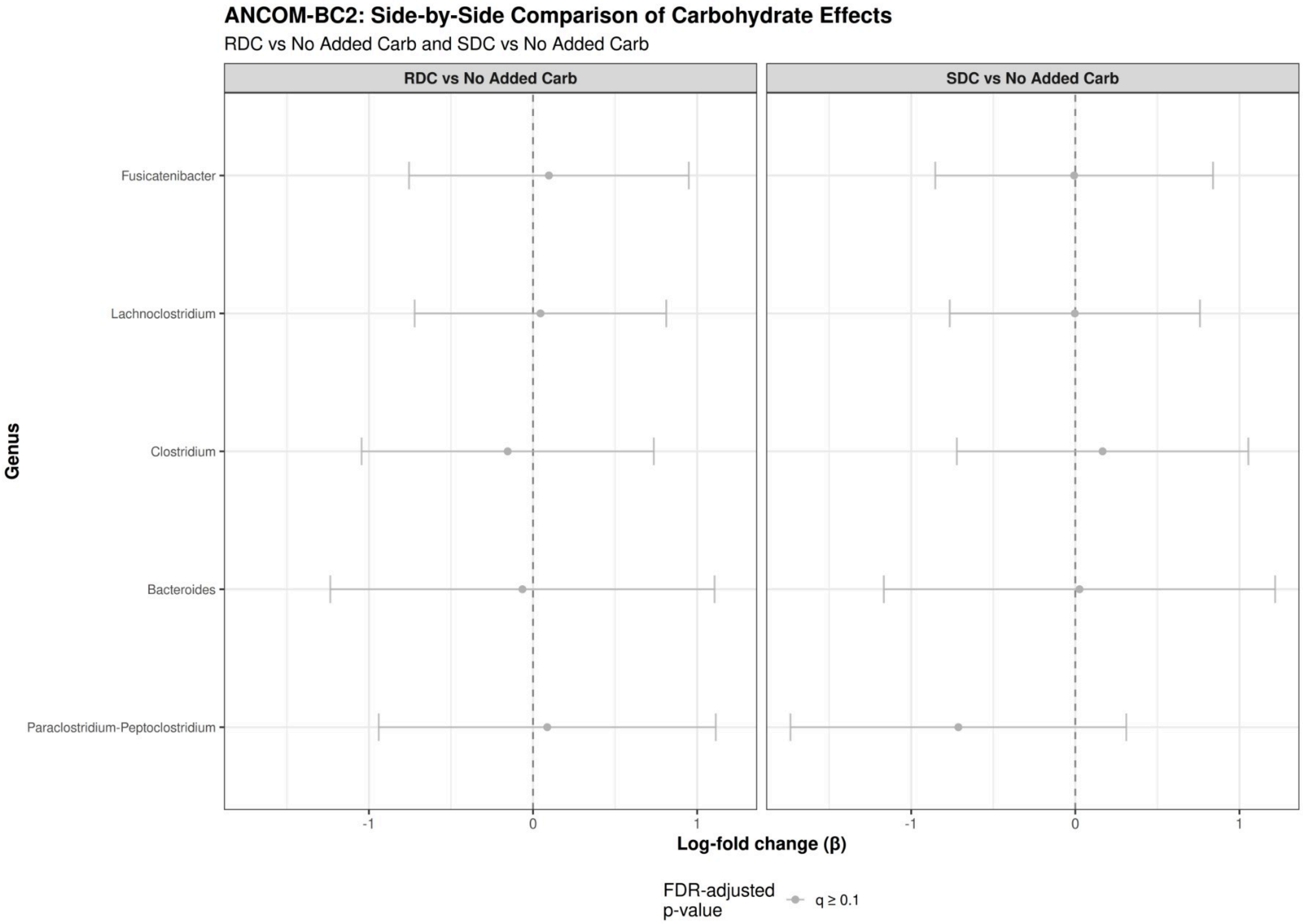
ANCOM-BC2 models microbial abundance using a generalized linear mixed-effects model with bias correction for compositional data. We tested the main effects of group (Healthy vs Obese), carbohydrate type (RDC vs No Added Carb, SDC vs No Added Carb), timepoint (48H vs 0H), and their interactions.

```
## ANCOM-BC2 significant results (q < 0.1): 10 taxon-effect combinations
```

6.5.2 ANCOM-BC2 Forest Plots



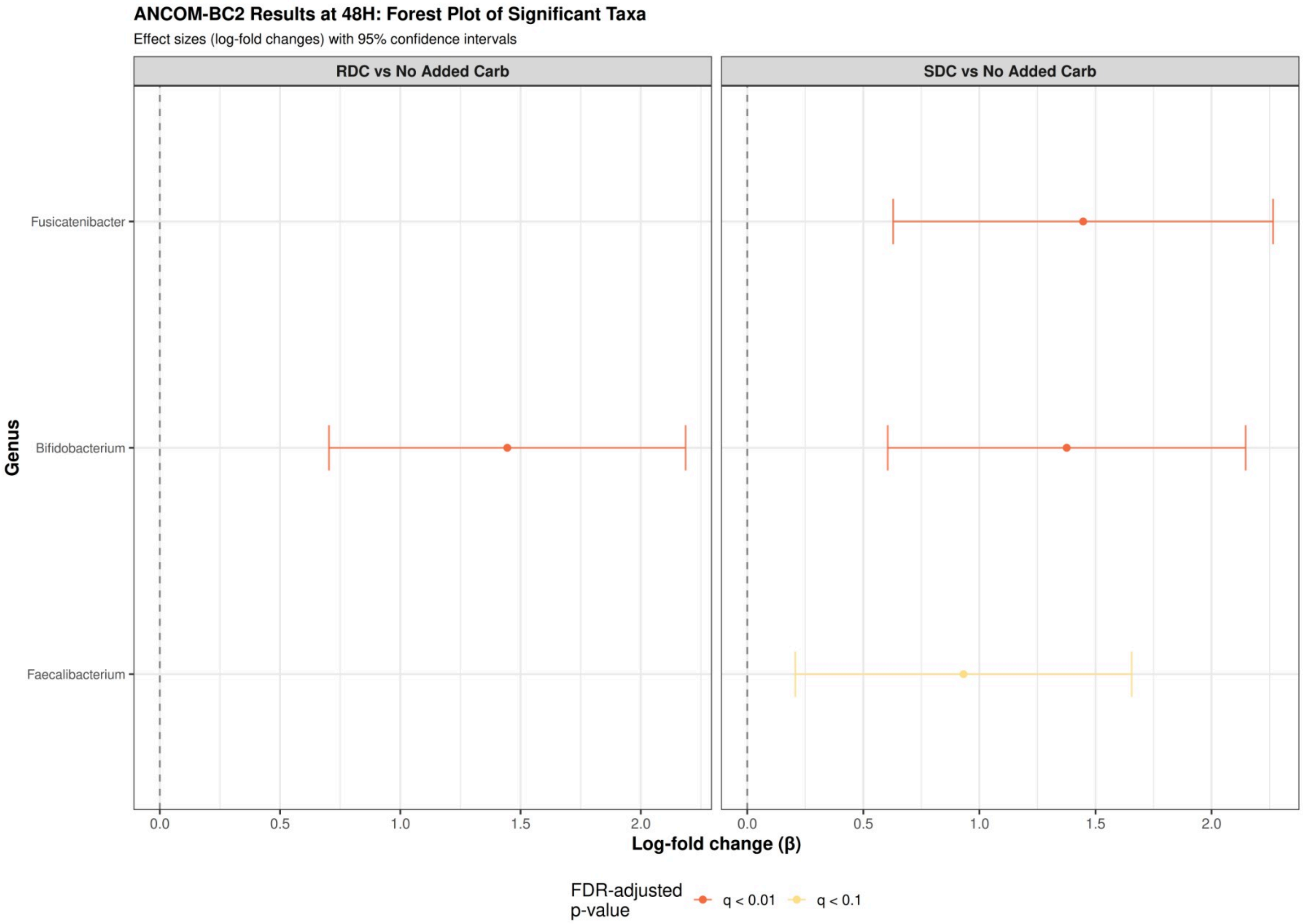
6.5.3 ANCOM-BC2 Side-by-Side Comparison: Carbohydrate Effects



6.5.4 ANCOM-BC2 48H Timepoint Analysis

```
## ANCOM-BC2 48H significant results (q < 0.1): 4 taxon-effect combinations
```

6.5.5 ANCOM-BC2 48H Forest Plots



6.5.6 ANCOM-BC2 RDC vs SDC at 48H

To directly compare the effects of RDC versus SDC at the 48H timepoint, we performed a focused ANCOM-BC2 analysis including only RDC and SDC samples at 48H, excluding the No Added Carb condition.

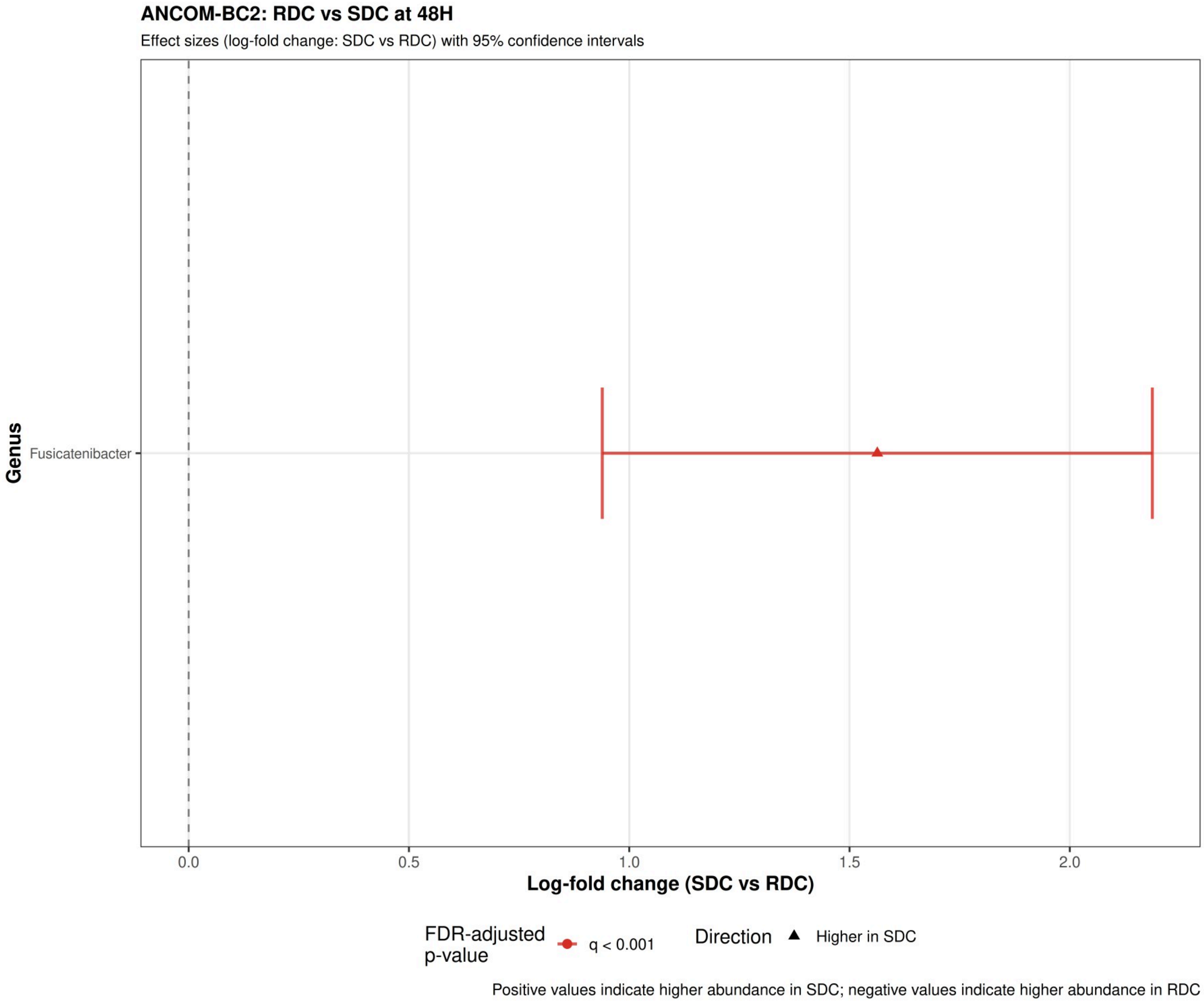
```
## 48H RDC vs SDC samples for ANCOM-BC2: 126
```

```
## Sample distribution by carbohydrate:
```

```
##
## RDC SDC
##  62  64
```

```
## ANCOM-BC2 48H RDC vs SDC significant results (q < 0.1): 1 taxa
```

6.5.7 Forest Plot: RDC vs SDC at 48H




```
##
## Summary of ANCOM-BC2 RDC vs SDC at 48H:
## Total taxa tested: 20
## Significant taxa (q < 0.1): 1
## Higher in SDC (q < 0.1): 1
## Higher in RDC (q < 0.1): 0
```

6.5.8 MaAsLin2 Analysis

MaAsLin2 (Microbiome Multivariable Associations with Linear Models) uses generalized linear models to identify associations between microbial taxa and metadata variables while accounting for confounding factors. We applied MaAsLin2 to identify taxa associated with group, carbohydrate type, and timepoint.

```
## MaAsLin2 significant results (q < 0.1): 32 feature-metadata combinations
```

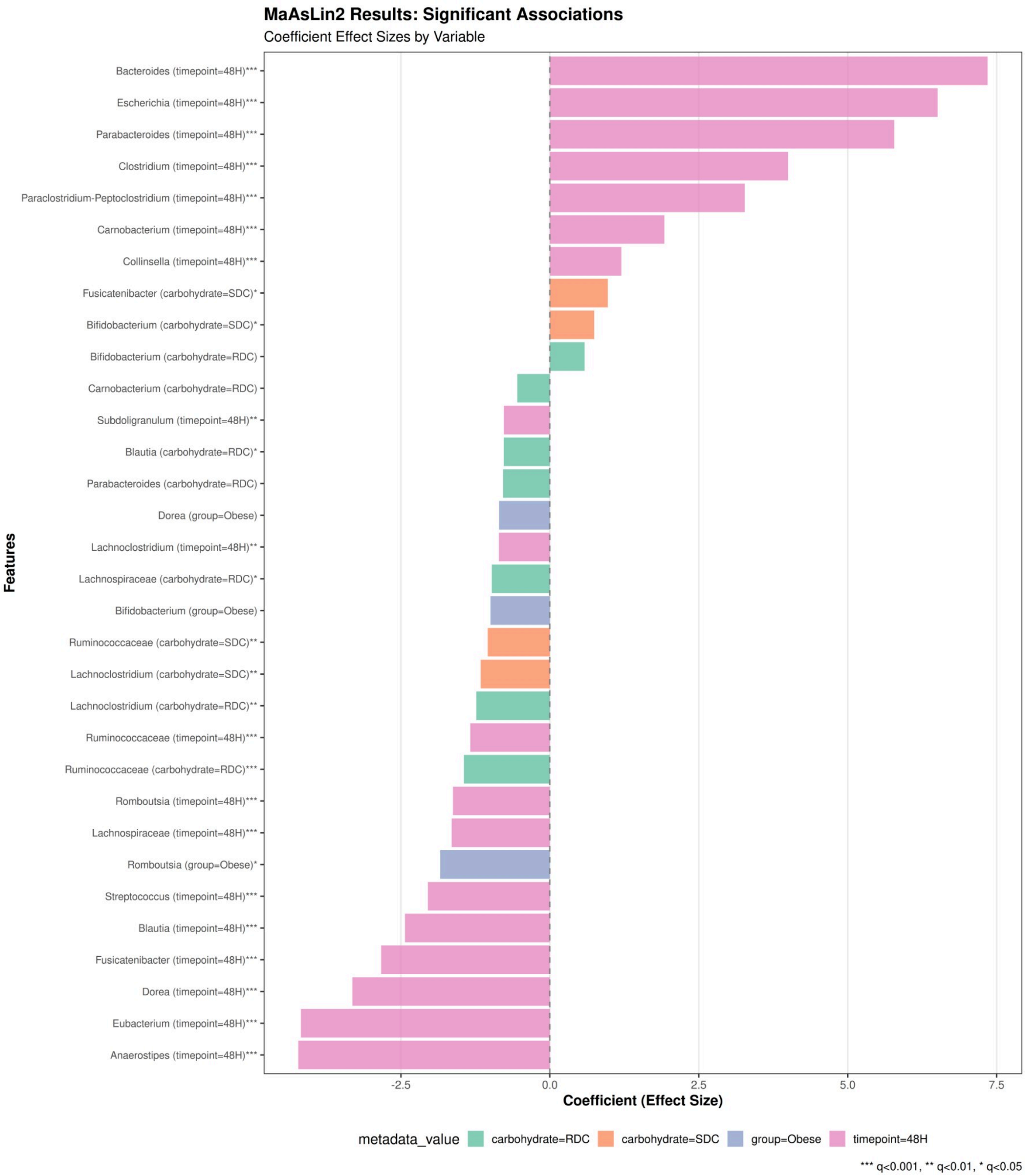
6.5.9 MaAsLin2 Summary Table

Summary of Differential Abundance Results (q < 0.1)

	taxon	Genus Species	DA Method	Variable	Effect Size (coef)	p value	
8	seq1	Blautia	MaAsLin2	timepoint=48H	-2.433	8.231999999999999e-29	9.408
9	seq1	Blautia	MaAsLin2	carbohydrate=RDC	-0.7752	0.004065	0.0
13	seq10	Clostridium	MaAsLin2	timepoint=48H	3.999	1.205e-27	1.205
14	seq10	Clostridium perfringens	ANCOM-BC2	timepoint48H	3.022	8.788e-7	0.00000
23	seq13	Fusicatenibacter	MaAsLin2	timepoint=48H	-2.833	3.953e-22	3.514
24	seq13	Fusicatenibacter	MaAsLin2	carbohydrate=SDC	0.9733	0.008001	0.0
25	seq13	Fusicatenibacter saccharivorans	ANCOM-BC2	timepoint48H	-1.399	0.02206	0.0
26	seq14	Lachnoclostridium	MaAsLin2	carbohydrate=RDC	-1.235	0.0003249	0.00
27	seq14	Lachnoclostridium	MaAsLin2	timepoint=48H	-0.8555	0.0005913	0.0
28	seq14	Lachnoclostridium	MaAsLin2	carbohydrate=SDC	-1.16	0.0007063	0.00
29	seq14	Lachnoclostridium sp32380-sp32437	ANCOM-BC2	timepoint48H	1.089	0.04496	0.0
10	seq15	Carnobacterium	MaAsLin2	timepoint=48H	1.923	7.530000000000001e-20	6.024
11	seq15	Carnobacterium	MaAsLin2	carbohydrate=RDC	-0.5482	0.0393	0.0
12	seq15	Carnobacterium mobile	ANCOM-BC2	timepoint48H	3.472	0.008223	0.0
16	seq17	Dorea	MaAsLin2	timepoint=48H	-3.314	6.163e-33	8.2170000000000
17	seq17	Dorea	MaAsLin2	group=Obese	-0.8518	0.02385	0.0
18	seq17	Dorea longicatena	ANCOM-BC2	timepoint48H	-1.672	0.001529	0.00
21	seq18	Eubacterium	MaAsLin2	timepoint=48H	-4.179	6.609e-40	1.057

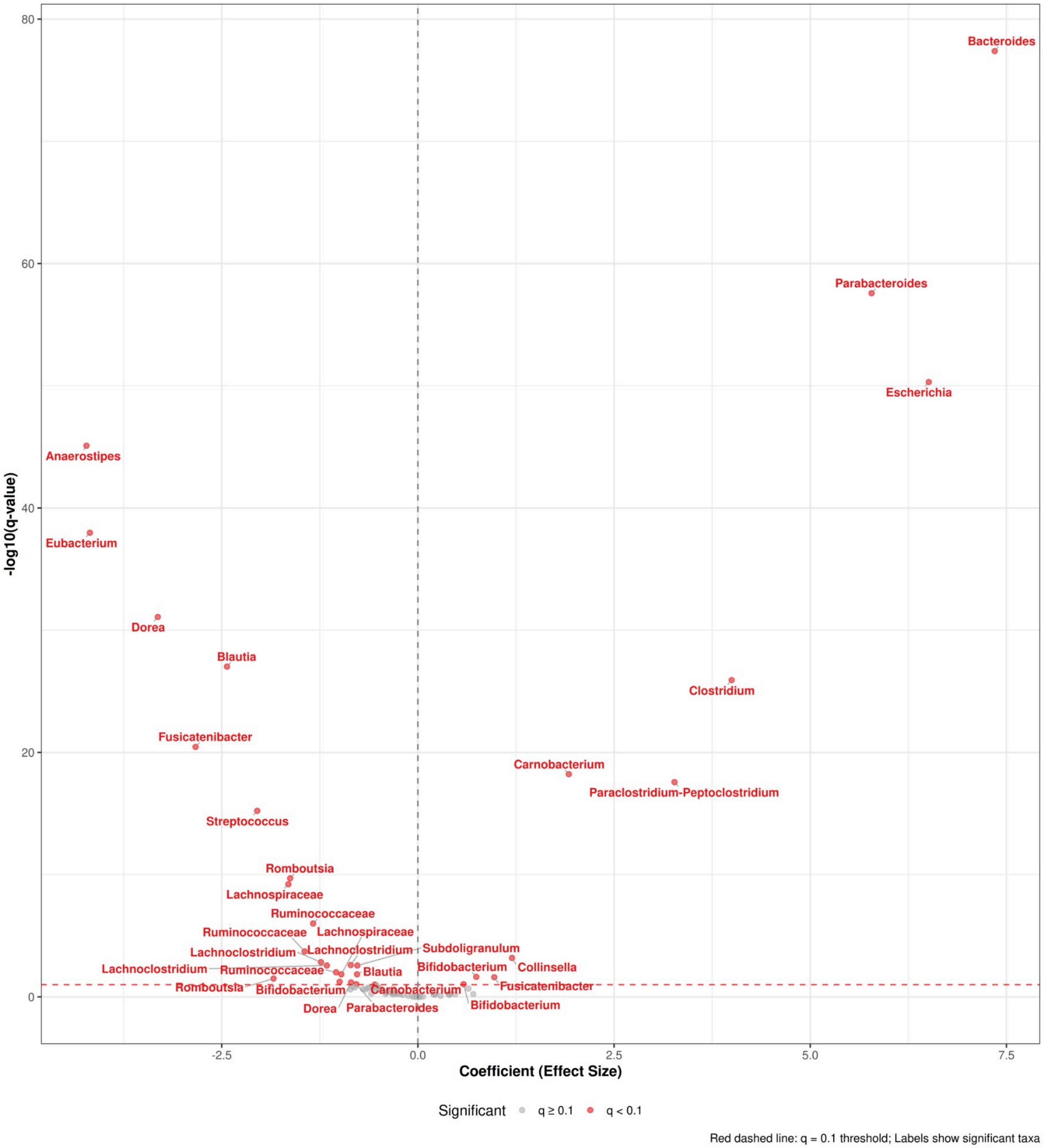
22	seq18	Eubacterium hallii	ANCOM-BC2	timepoint48H	-2.226	0.00003277	0.000
19	seq2	Escherichia	MaAsLin2	timepoint=48H	6.511	1.861e-52	4.962999999999
20	seq2	Escherichia coli	ANCOM-BC2	timepoint48H	2.907	0.005243	0.0
41	seq20	Streptococcus	MaAsLin2	timepoint=48H	-2.048	8.996e-17	5.99
42	seq21	Subdoligranulum	MaAsLin2	timepoint=48H	-0.7719	0.0007085	0.00
30	seq29	Lachnospiraceae	MaAsLin2	timepoint=48H	-1.65	1.059e-10	6.05
31	seq29	Lachnospiraceae	MaAsLin2	carbohydrate=RDC	-0.9757	0.004255	0.0
5	seq3	Bifidobacterium	MaAsLin2	carbohydrate=SDC	0.7446	0.007189	(
6	seq3	Bifidobacterium	MaAsLin2	group=Obese	-0.9972	0.02035	0.0
7	seq3	Bifidobacterium	MaAsLin2	carbohydrate=RDC	0.5824	0.03543	0.0
3	seq31	Bacteroides	MaAsLin2	timepoint=48H	7.351	5.210999999999999e-80	4.16
4	seq31	Bacteroides vulgatus	ANCOM-BC2	timepoint48H	4.283	7.28e-9	1.4
1	seq34	Anaerostipes	MaAsLin2	timepoint=48H	-4.223	4.006e-47	8.011
2	seq34	Anaerostipes hadrus	ANCOM-BC2	timepoint48H	-1.679	0.002213	0.00
15	seq4	Collinsella	MaAsLin2	timepoint=48H	1.199	0.0001378	0.000
38	seq41	Ruminococcaceae	MaAsLin2	timepoint=48H	-1.335	1.861e-7	9.9
39	seq41	Ruminococcaceae	MaAsLin2	carbohydrate=RDC	-1.443	0.00003834	0.000
40	seq41	Ruminococcaceae	MaAsLin2	carbohydrate=SDC	-1.043	0.002672	0.00
32	seq46	Parabacteroides	MaAsLin2	timepoint=48H	5.782	6.675999999999999e-60	2.6
33	seq46	Parabacteroides	MaAsLin2	carbohydrate=RDC	-0.7848	0.03444	0.0
34	seq46	Parabacteroides distasonis	ANCOM-BC2	timepoint48H	3.898	0.000004284	0.0000
35	seq6	Paraclostridium-Peptoclostridium	MaAsLin2	timepoint=48H	3.27	3.686e-19	2.681
36	seq8	Romboutsia	MaAsLin2	timepoint=48H	-1.627	3.214e-11	1.97
37	seq8	Romboutsia	MaAsLin2	group=Obese	-1.84	0.01073	0.0

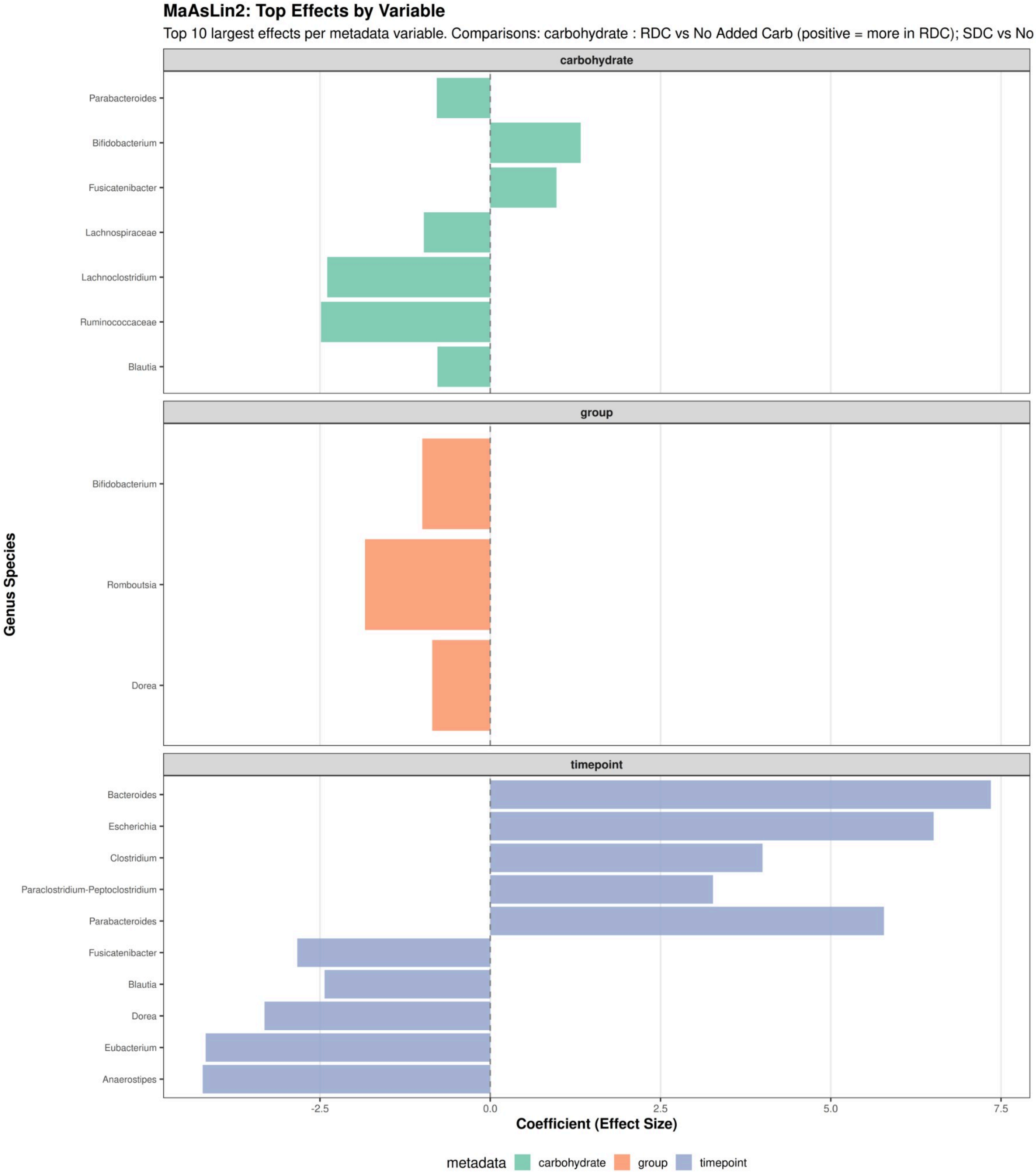
6.5.10 MaAsLin2 Visualizations



MaAsLin2 Volcano Plot

Effect Size vs Statistical Significance

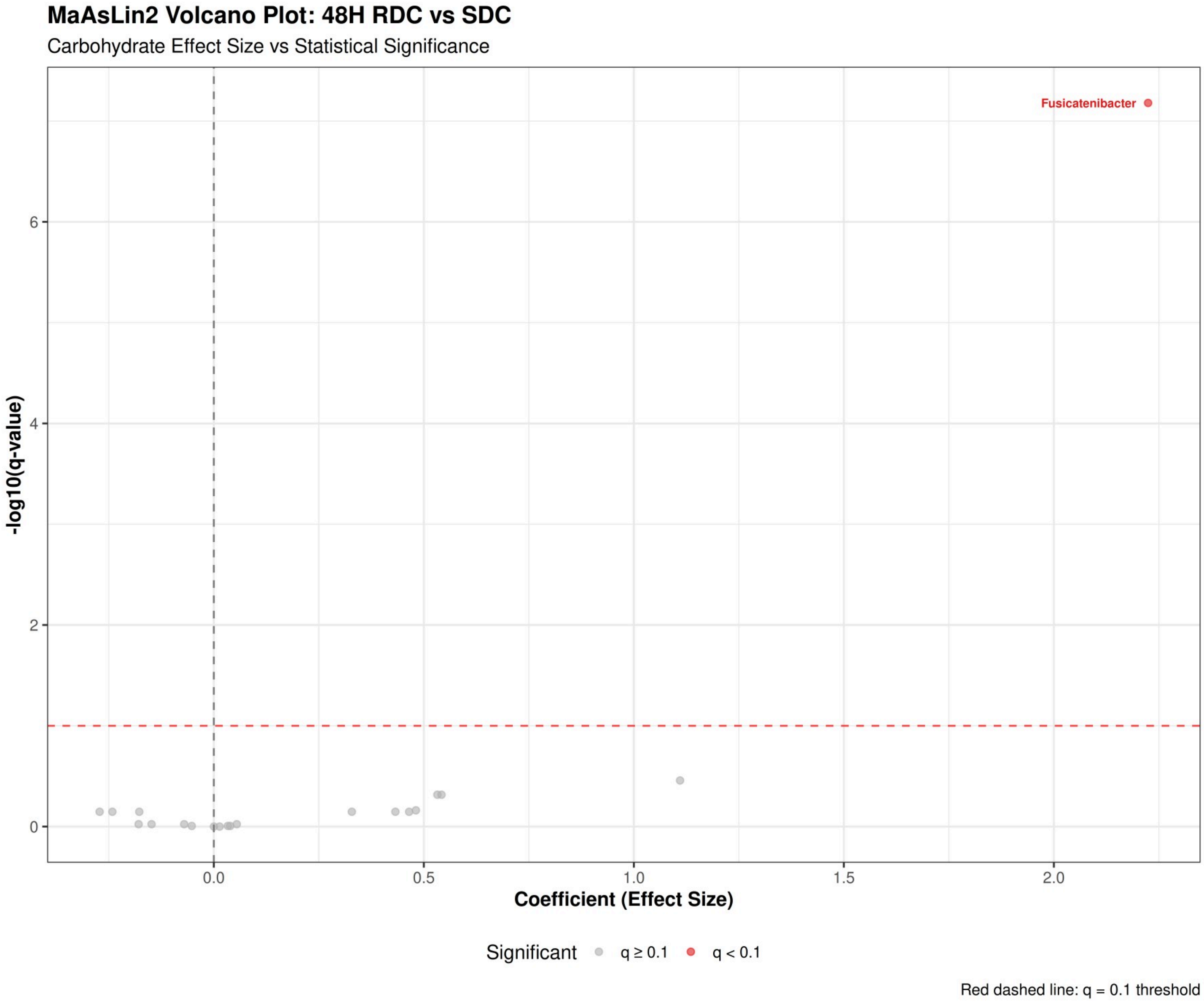




6.5.11 MaAsLin2 48H RDC vs SDC Analysis

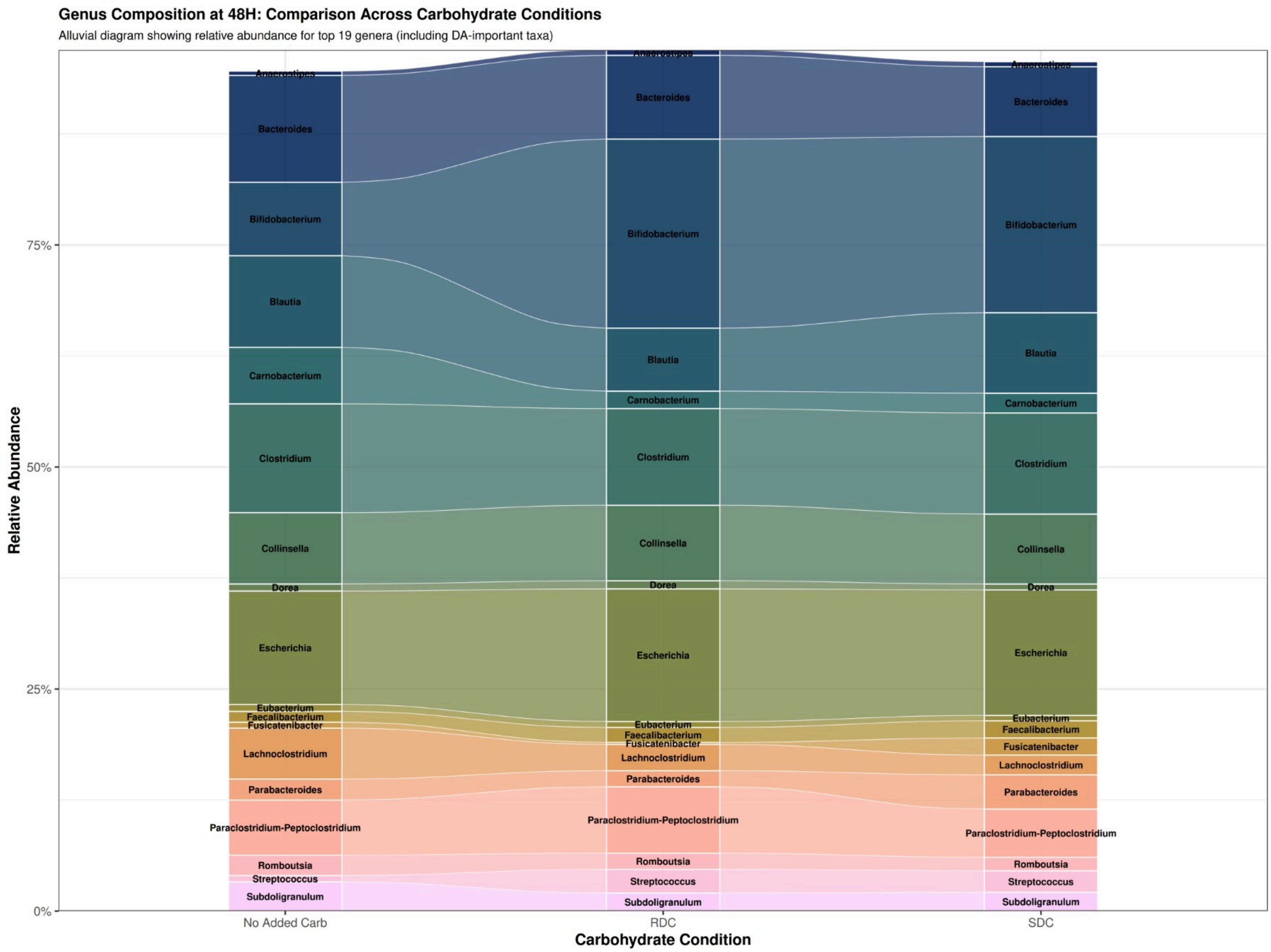
```
## MaAsLin2 48H RDC vs SDC significant results (q < 0.1): 1
```

6.5.12 MaAsLin2 48H RDC vs SDC Volcano Plot



6.4.4 Genus Composition at 48H: No Carbohydrate vs RDC vs SDC

This alluvial plot shows the comparison of genus-level composition at 48H across the three carbohydrate conditions (No Added Carb, RDC, and SDC), highlighting the top 25 most abundant genera and including key differentially abundant taxa.




```
##
## 48H Carbohydrate Comparison Summary:
```

```
## - Total genera included: 19
```

```
## - DA-important genera found: 1
```

```
## - DA-important genera: g__Fusicatenibacter
```

6.6 Absolute Abundances

Relative abundance data (compositional) can mask true changes in microbial load. To address this, we integrated qPCR-derived total genome copy estimates per sample to convert relative abundances to absolute abundances (genome copies per sample). This allows us to distinguish between compositional shifts versus true changes in microbial load.

6.6.1 Integration of qPCR Data

```
## qPCR data integration:
```

```
## - Total samples in phyloseq: 292
```

```
## - Samples matched to qPCR data: 292
```

```
## - Match rate: 100 %
```

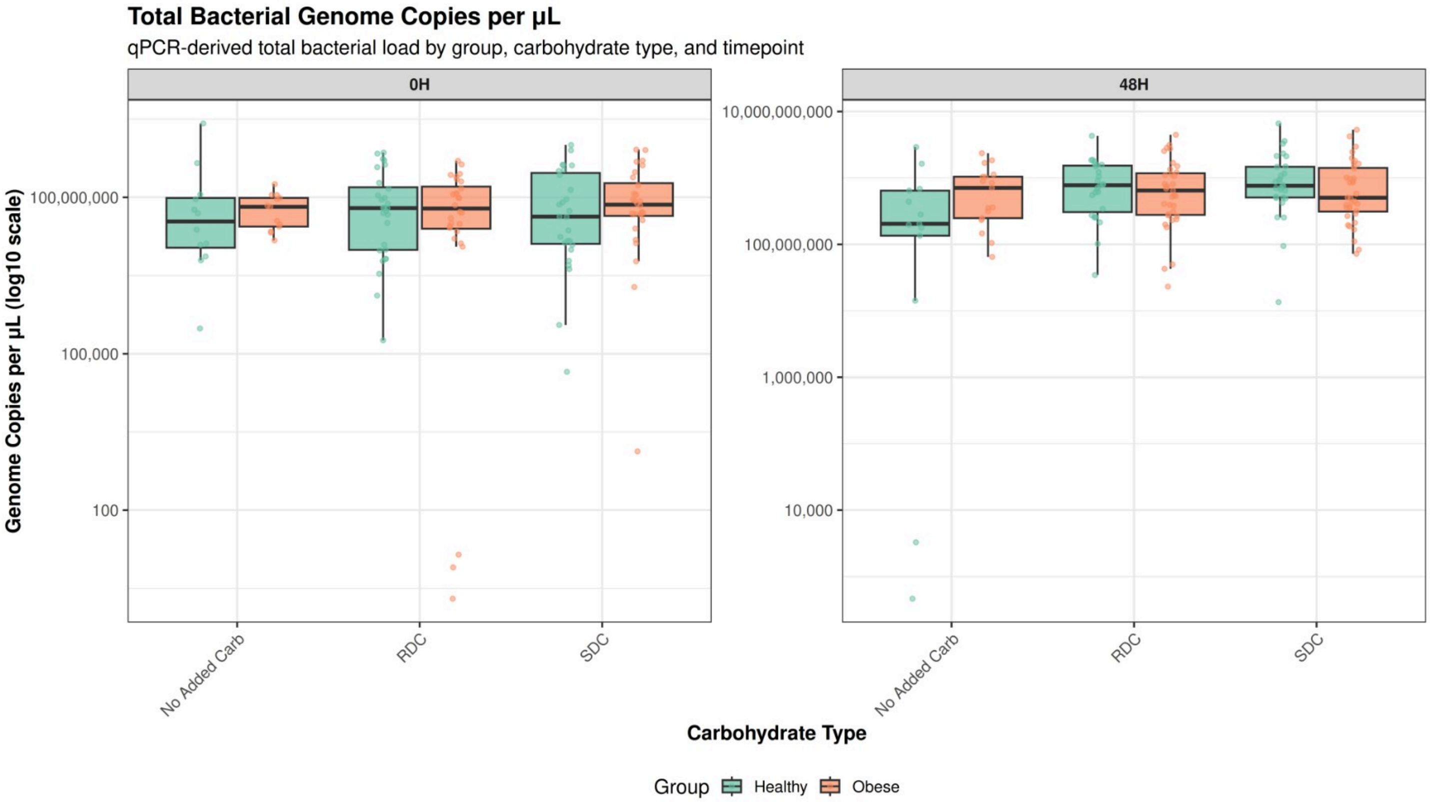
6.6.2 Conversion to Absolute Abundances

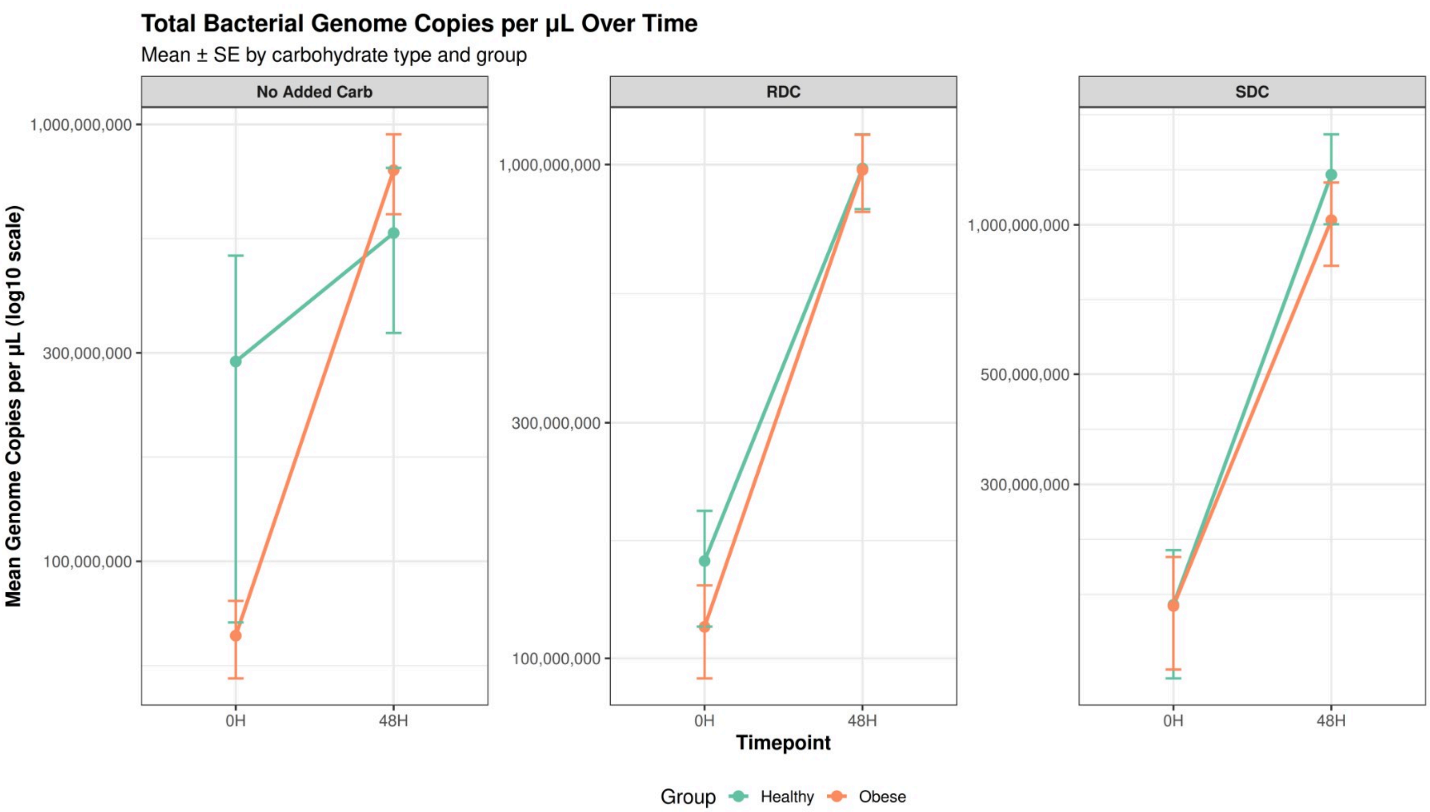
```
## Absolute abundance conversion complete:
```

```
## - Samples with qPCR data: 292
```

```
## - Samples without qPCR data: 0
```

6.6.3 Absolute Abundance Plots by Group, Carbohydrate, and Timepoint






```
##
## qPCR Data Summary:
```

```
## - Total samples with qPCR data: 292
```

```
## - Samples by timepoint:
```

```
##
##  0H 48H
## 136 156
```

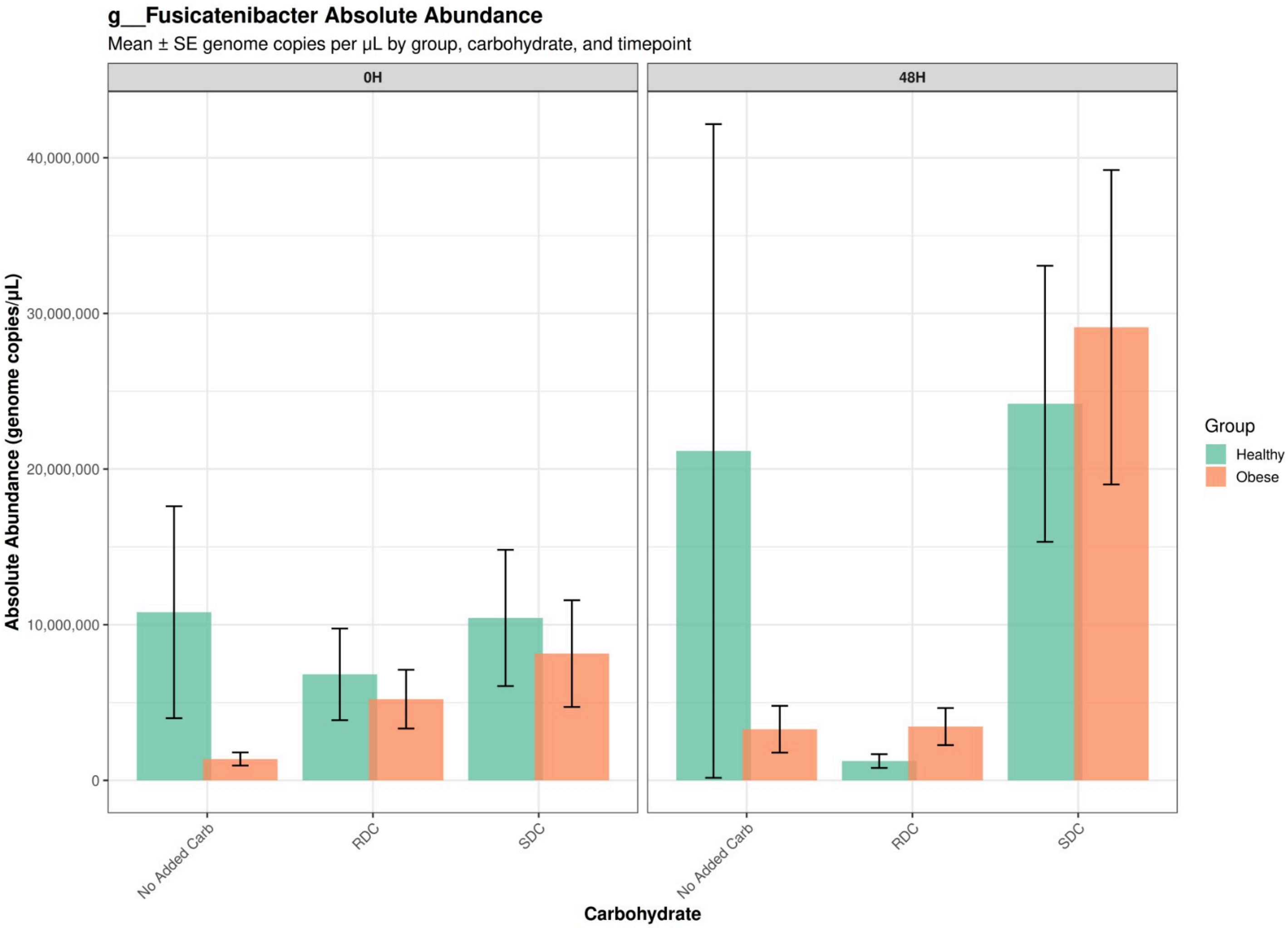
```
## - Samples by carbohydrate:
```

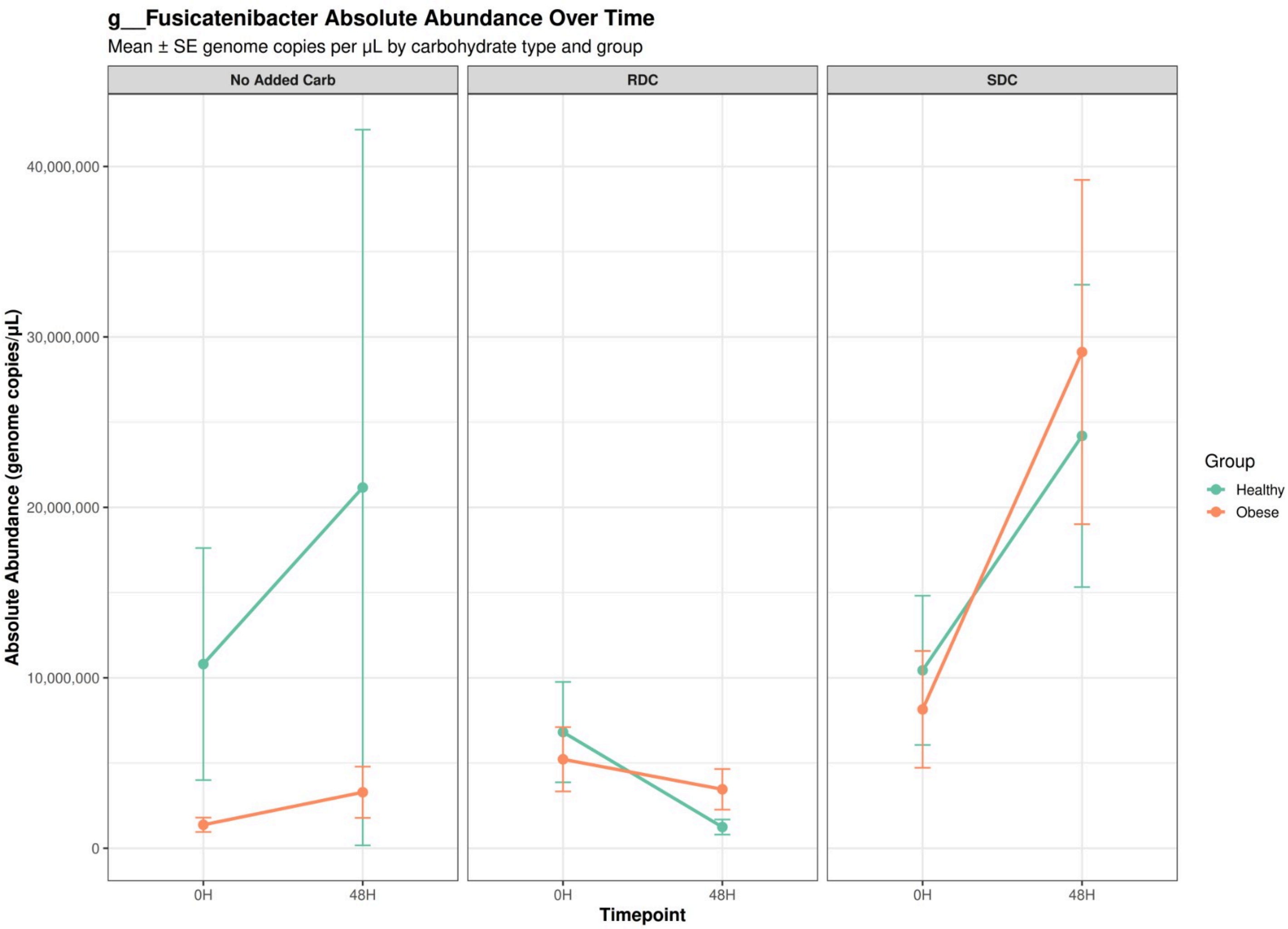
```
##
## No Added Carb           RDC           SDC
##            55           118           119
```

```
## - Samples by group:
```

```
##
## Healthy   Obese
##     135     157
```

6.6.4 Fusicatenibacter Absolute Abundance Plots





7. Integrated Analysis: Taxon–SCFA Relationships

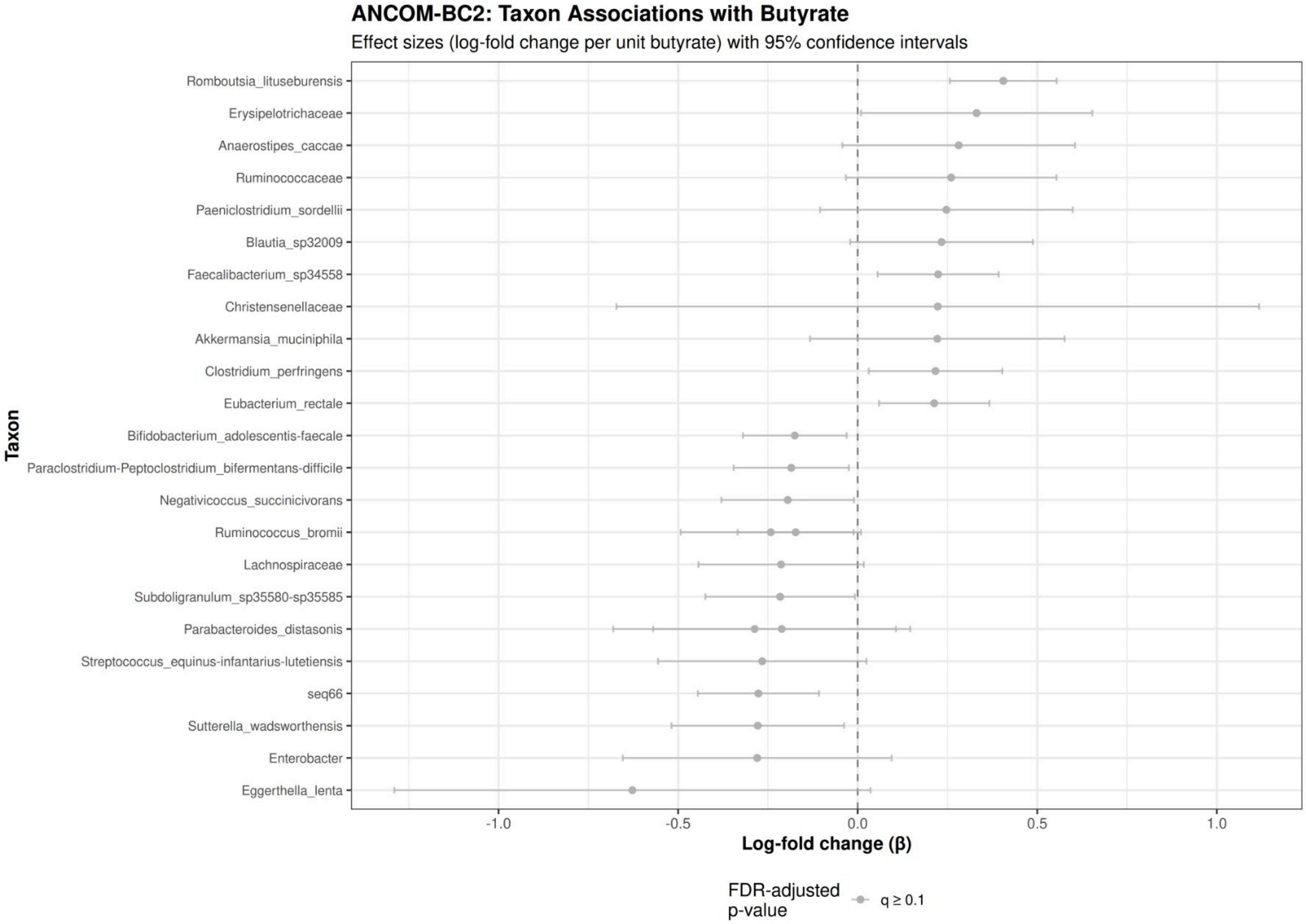
This section integrates SCFA metabolomics with microbiome composition to identify taxa associated with SCFA concentrations. We model associations between microbial taxa and butyrate/propionate concentrations while accounting for carbohydrate type effects, using both relative and absolute abundance data.

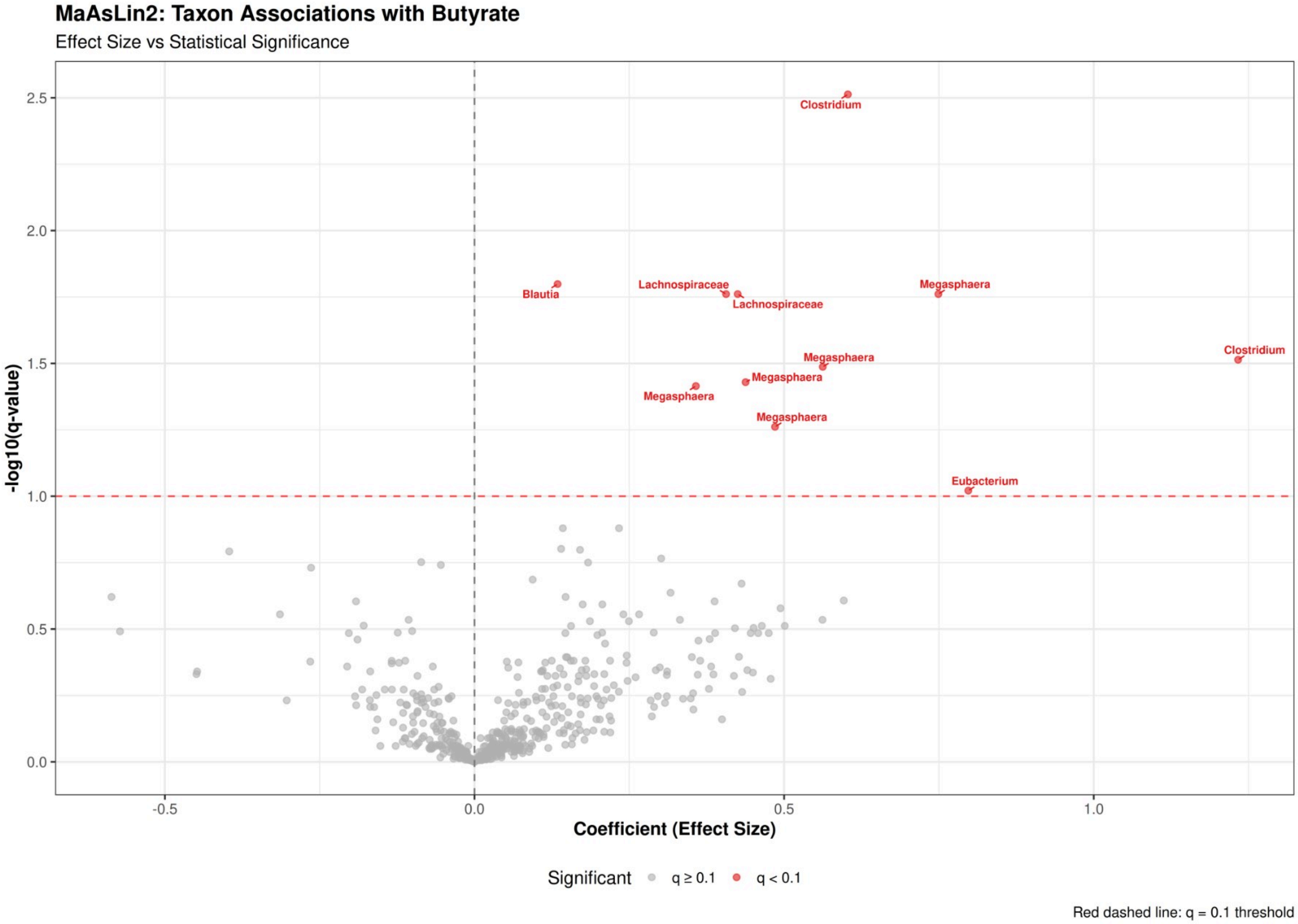
7.1 Integration of SCFA Data with Microbiome Metadata

7.2 Taxon–Butyrate Associations

7.2.1 ANCOM-BC2 and MaAsLin2 Visualizations: Butyrate

Associations

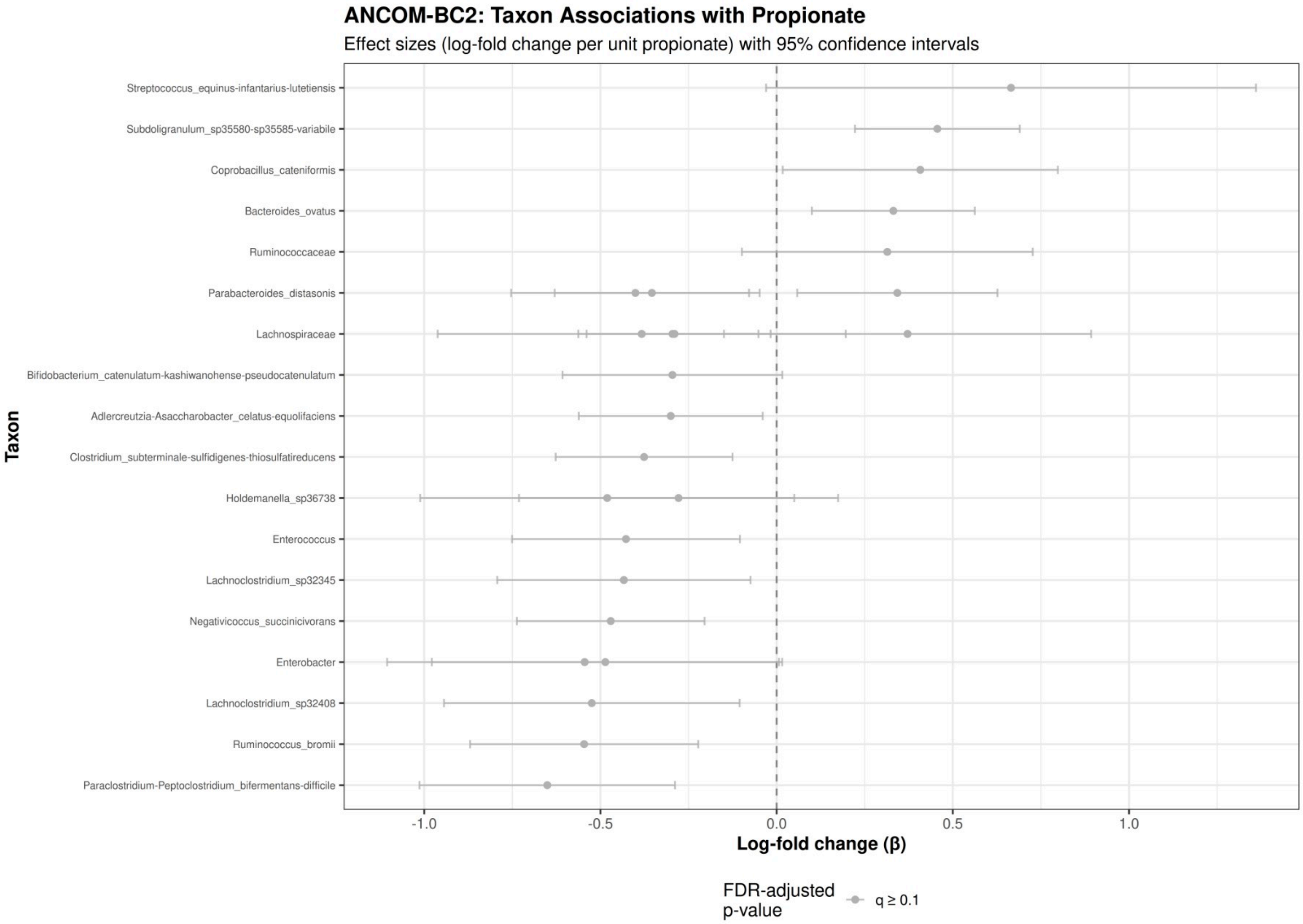


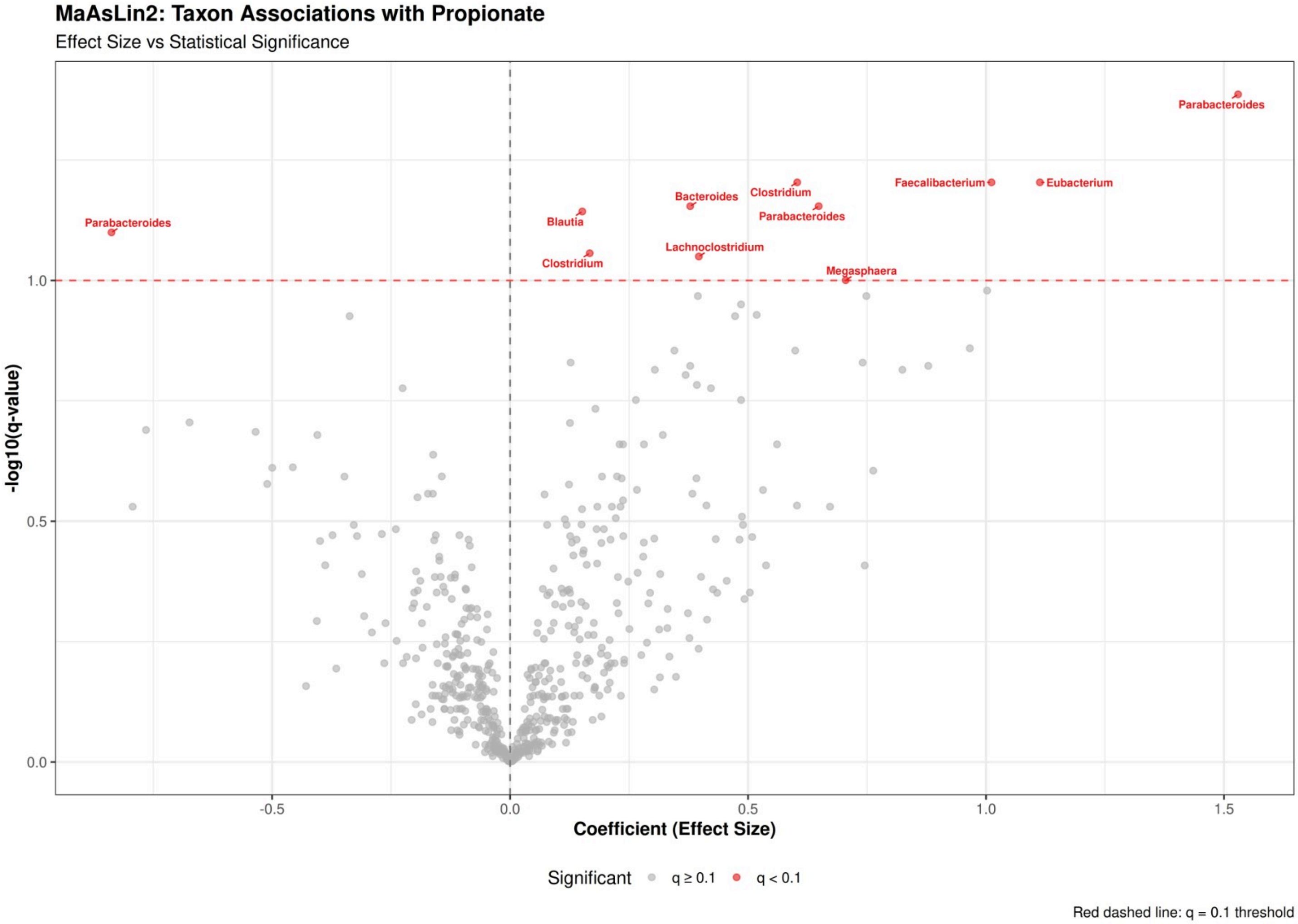


7.3 Taxon–Propionate Associations

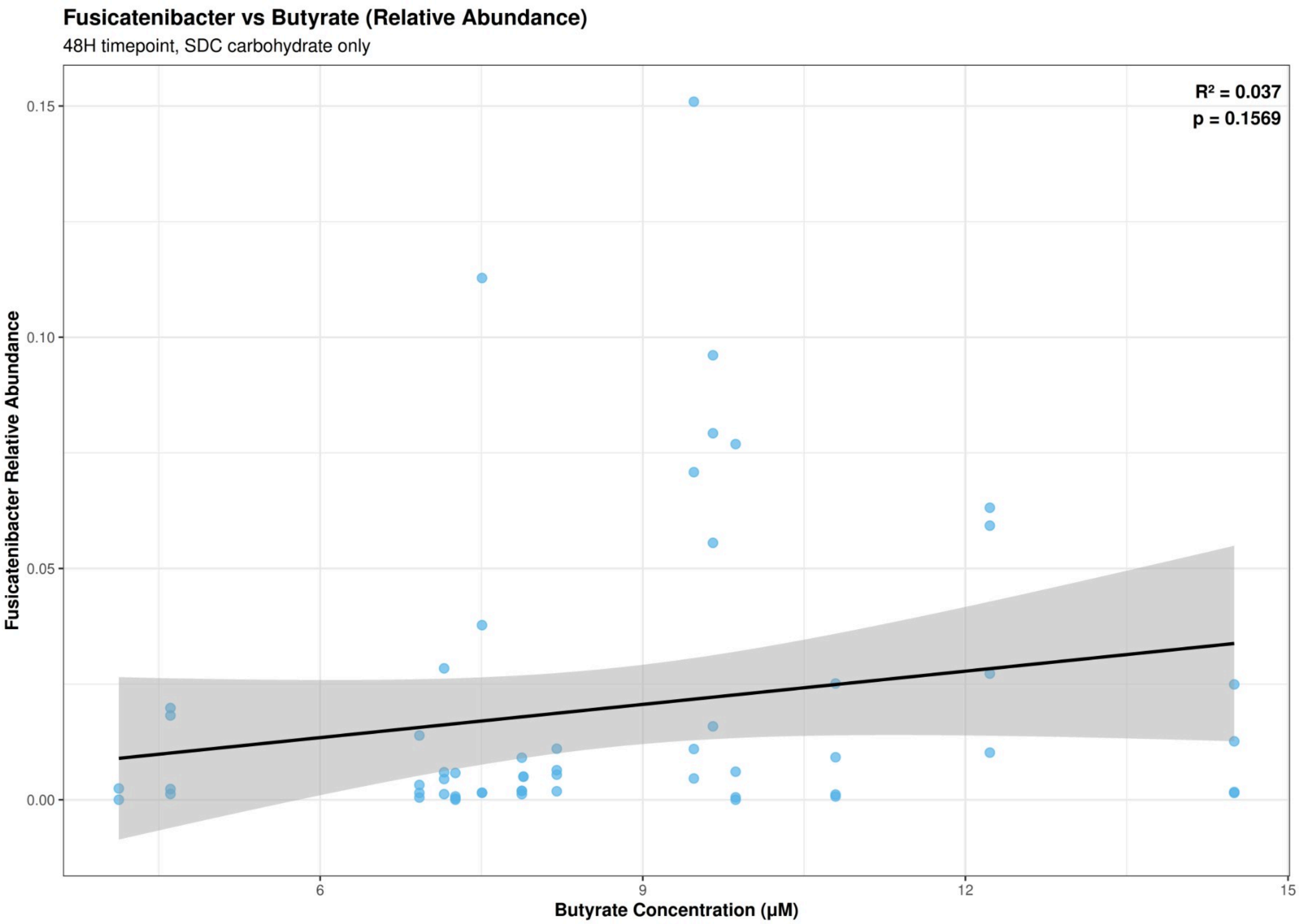
7.3.1 ANCOM-BC2 and MaAsLin2 Visualizations: Propionate

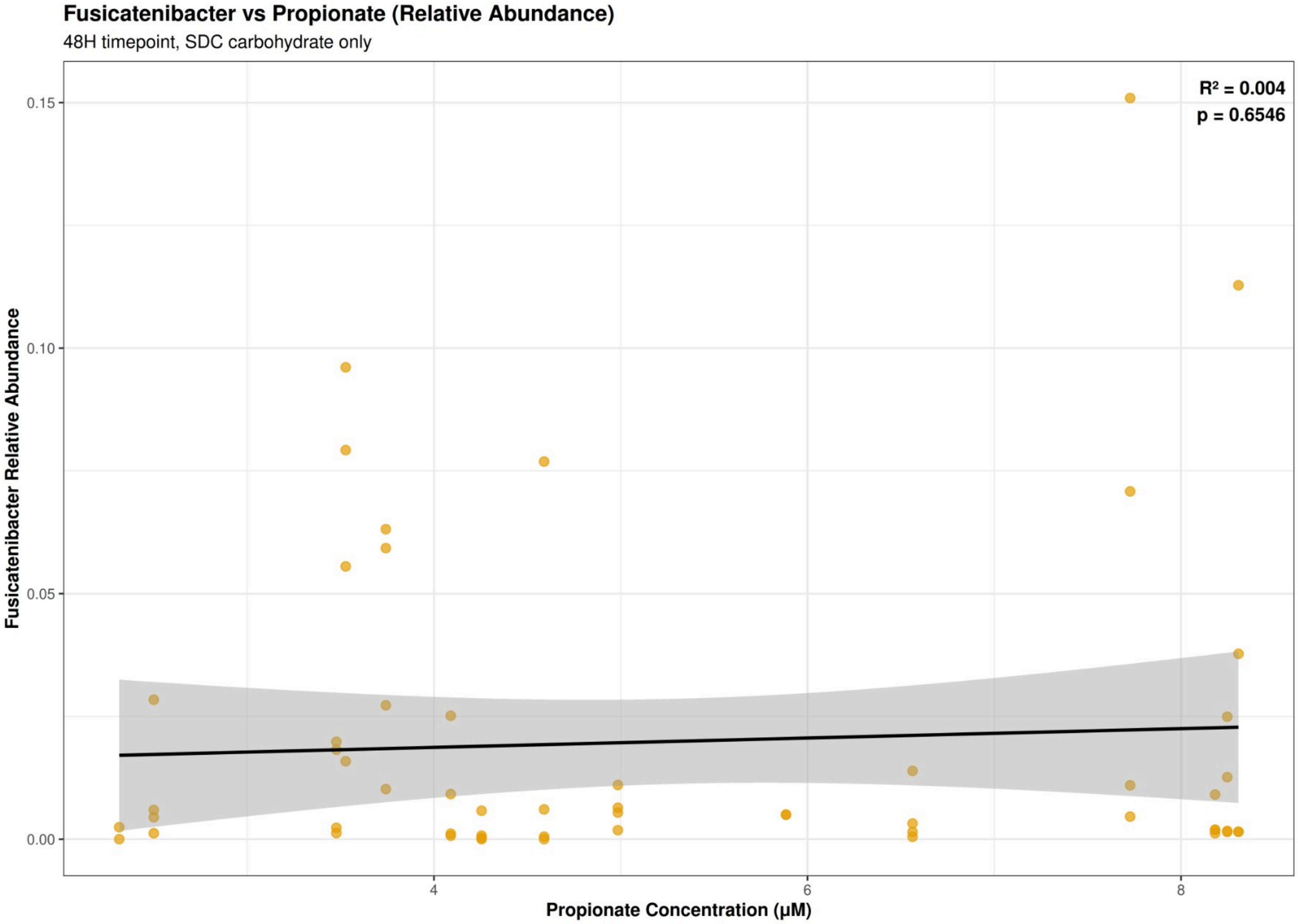
Associations




MaAsLin2: Taxon Associations with Propionate
Effect Size vs Statistical Significance
Parabacteroides
Faecalibacterium
Eubacterium
Clostridium
Bacteroides
Parabacteroides
Blautia
Parabacteroides
Lachnoclostridium
Clostridium
Megasphaera
-log10(q-value)
1.0
0.5
0.0
-0.5
0.0
0.5
1.0
1.5
Coefficient (Effect Size)
Significant
q ≥ 0.1
q < 0.1
Red dashed line: q = 0.1 threshold

7.4 Fusicatenibacter–SCFA Correlations






```
##
## Fusicatenibacter-SCFA Correlations (48H, SDC only):
## ==================================================
## Butyrate:
##   R² = 0.0368, p = 0.1569
##   Spearman rho = 0.287
##   N = 56
##
## Propionate:
##   R² = 0.0037, p = 0.6546
##   Spearman rho = -0.031
##   N = 56
```

7.5 Responder vs Non-Responder Analysis

```
## Responder Status Overlap:
```

```
## - Total subjects: 48
```

```
##                 Propionate
## Butyrate        Non-Responder Responder
##   Non-Responder            12        12
##   Responder                12        12
## - Both Responder: 12 (25.0%)
## - Both Non-Responder: 12 (25.0%)
```

```
##
## 48H Microbiome Composition Comparison:
```

```
## - Samples with responder status: 136
```

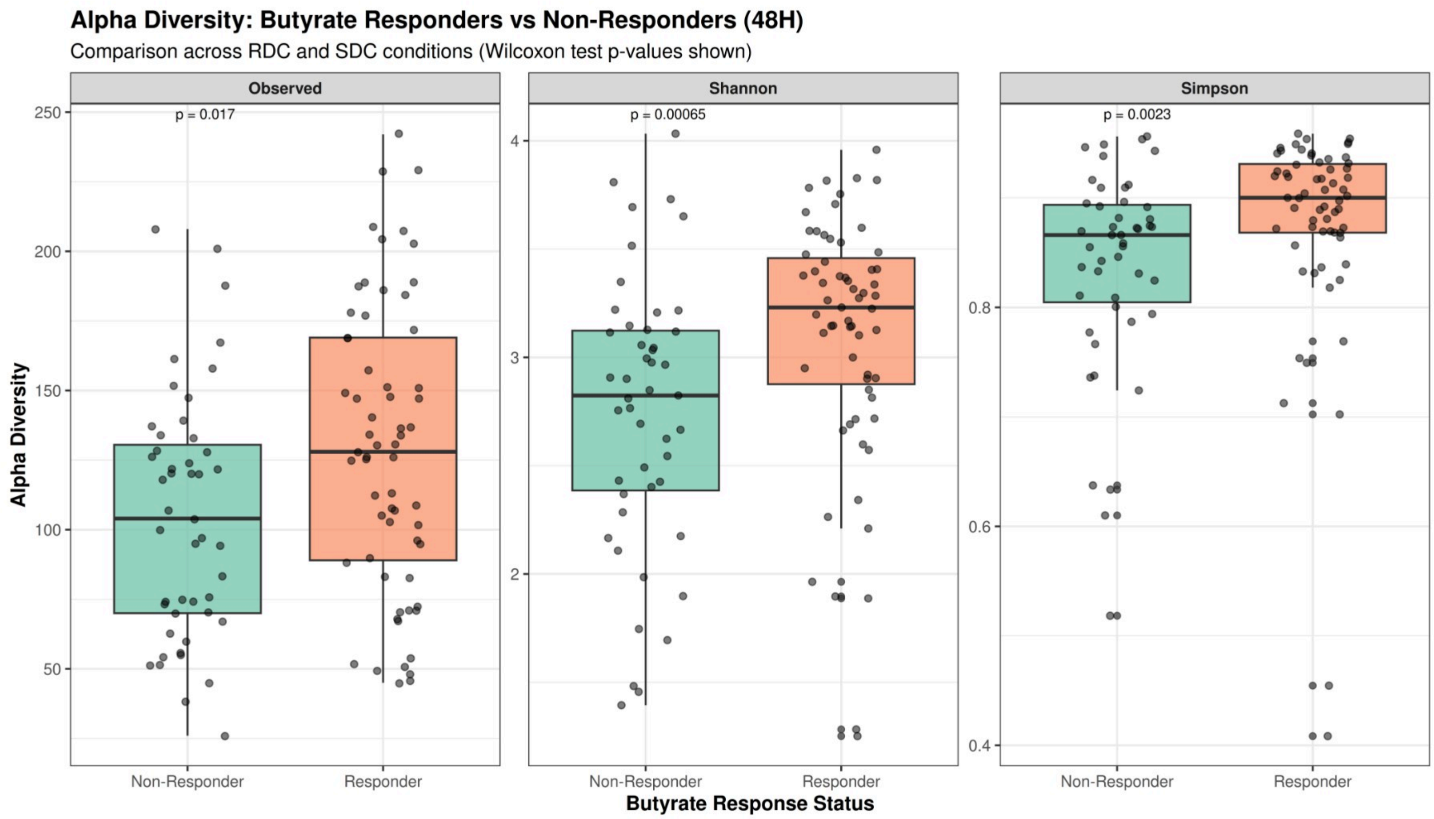
7.6 Microbial Composition Differences: Butyrate Responders vs Non-Responders

To investigate whether individual variation in SCFA response is associated with differences in microbiome composition, we classified subjects as "Responders" or "Non-Responders" based on their change in SCFA concentration from baseline (0H) to 48 hours (48H).

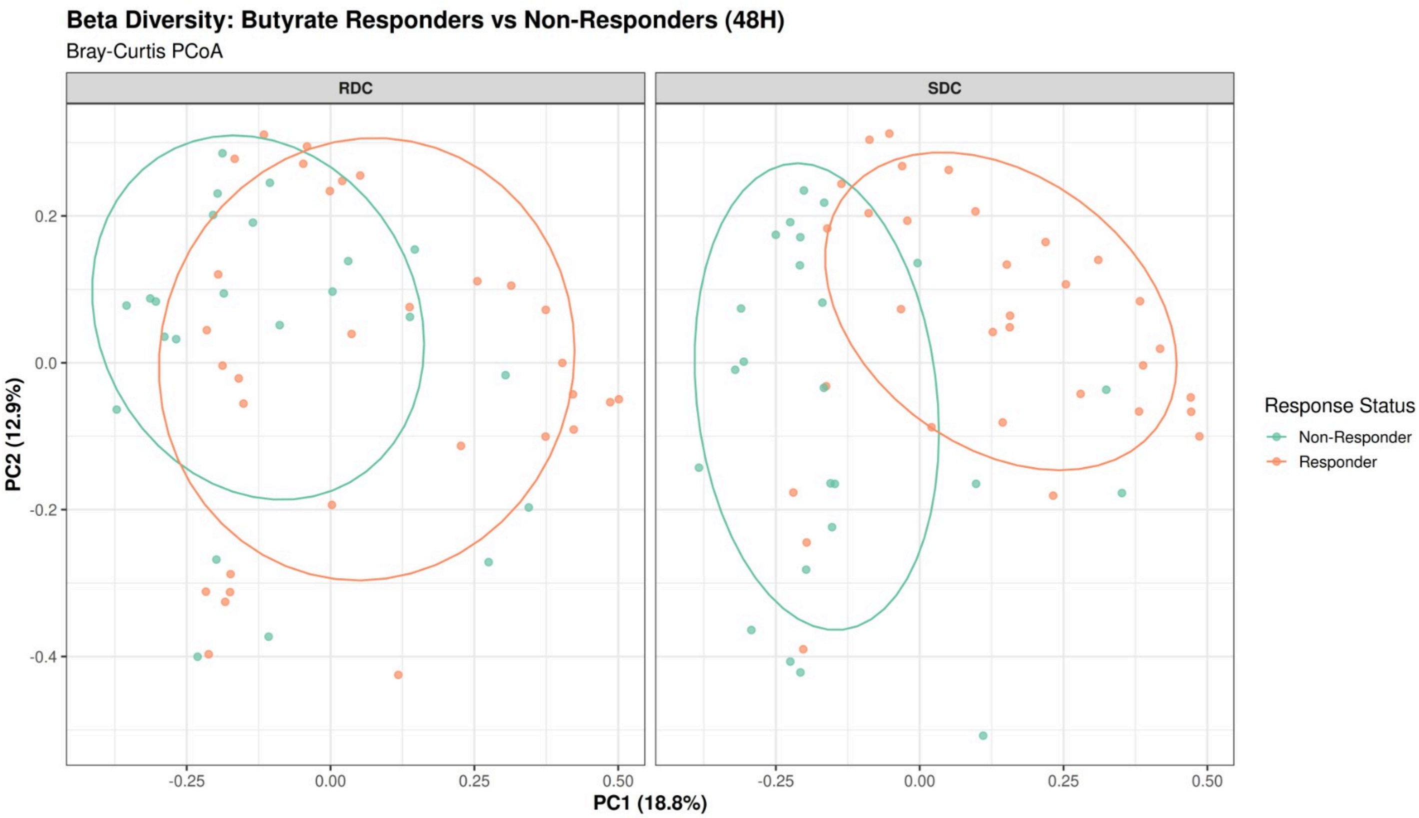
Definition of Responder Status: - For each SCFA (butyrate and propionate) and each carbohydrate condition (RDC and SDC), we calculated the delta change: $\Delta = [SCFA]_{48h} - [SCFA]_{0h}$ - Subjects were classified separately for each analyte and carbohydrate type using a median split approach - **Responders**: Subjects with $\Delta \geq$ median Δ for that specific analyte-carbohydrate combination - **Non-Responders**: Subjects with $\Delta <$ median Δ for that specific analyte-carbohydrate combination

This approach allows us to identify whether individuals who show greater increases in SCFA production (responders) have distinct microbial communities compared to those with smaller increases (non-responders), potentially revealing microbiome features associated with enhanced SCFA production capacity.

In this section, we compare the microbial composition (alpha diversity, beta diversity, and community structure) between butyrate responders and non-responders at the 48H timepoint, focusing on samples from RDC and SDC conditions (excluding No Added Carb controls).




```
##
## Wilcoxon Test Results: Butyrate Responders vs Non-Responders (Alpha Diversity)
## ==========================================================================
## # A tibble: 3 × 6
##   Metric   Group1        Group2    `p value` `p adjusted` Significance
##   <chr>    <chr>         <chr>         <dbl>        <dbl> <chr>
## 1 Observed Non-Responder Responder  0.0167        0.0167  *
## 2 Shannon  Non-Responder Responder  0.000655      0.00196 **
## 3 Simpson  Non-Responder Responder  0.00232       0.00348 **
```




```
##
## PERMANOVA: Butyrate Responder Status (48H):
## Permutation test for adonis under reduced model
## Blocks:  strata
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = dist_bray_resp ~ responder_butyrate * carbohydrate, data = me
ta_resp, permutations = 999, strata = meta_resp$subject_id)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      3    2.835 0.08264 3.1832  0.017 *
## Residual 106   31.473 0.91736
## Total    109   34.308 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

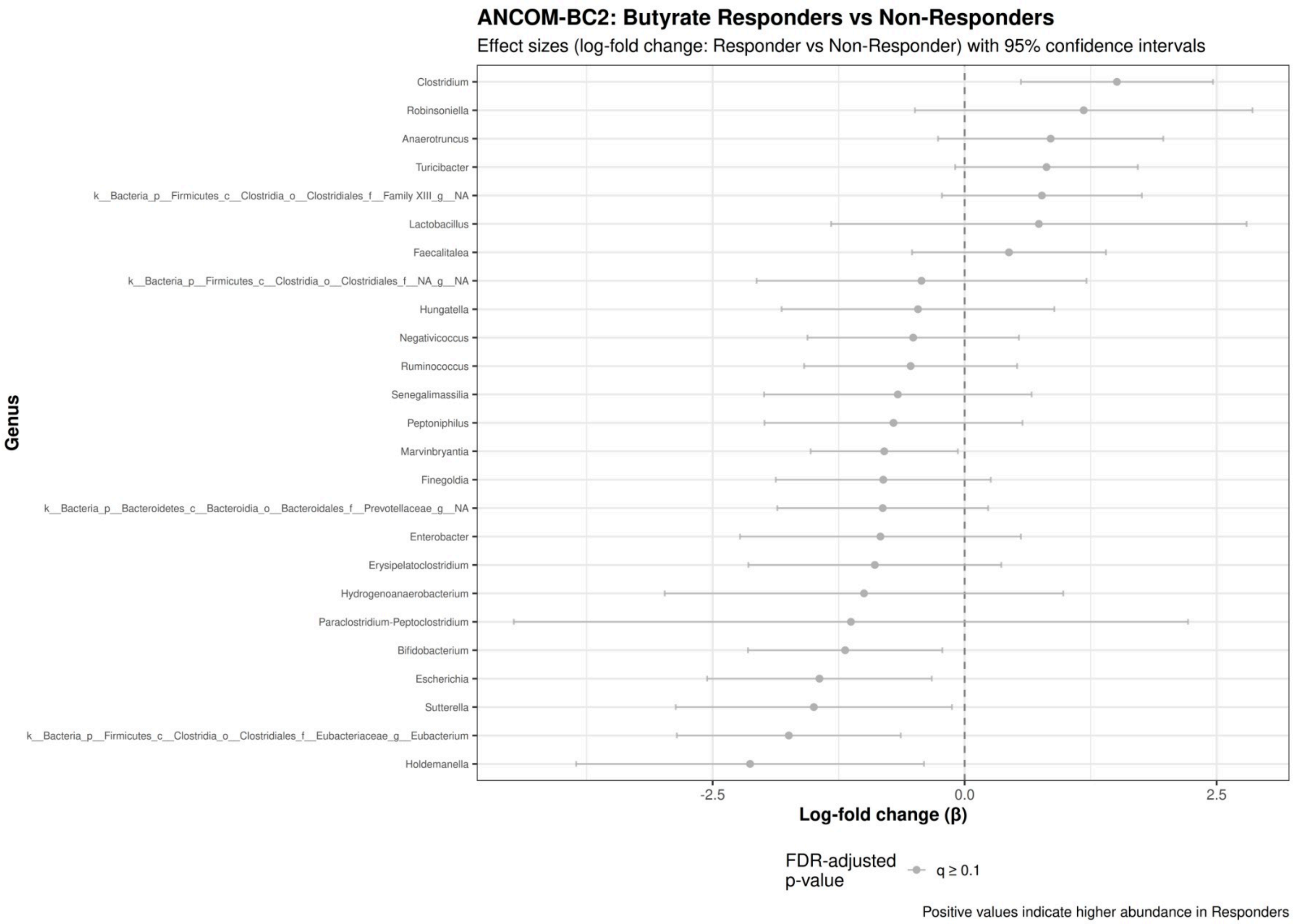
7.6.1 Differential Abundance Analysis: Butyrate Responders vs Non-Responders

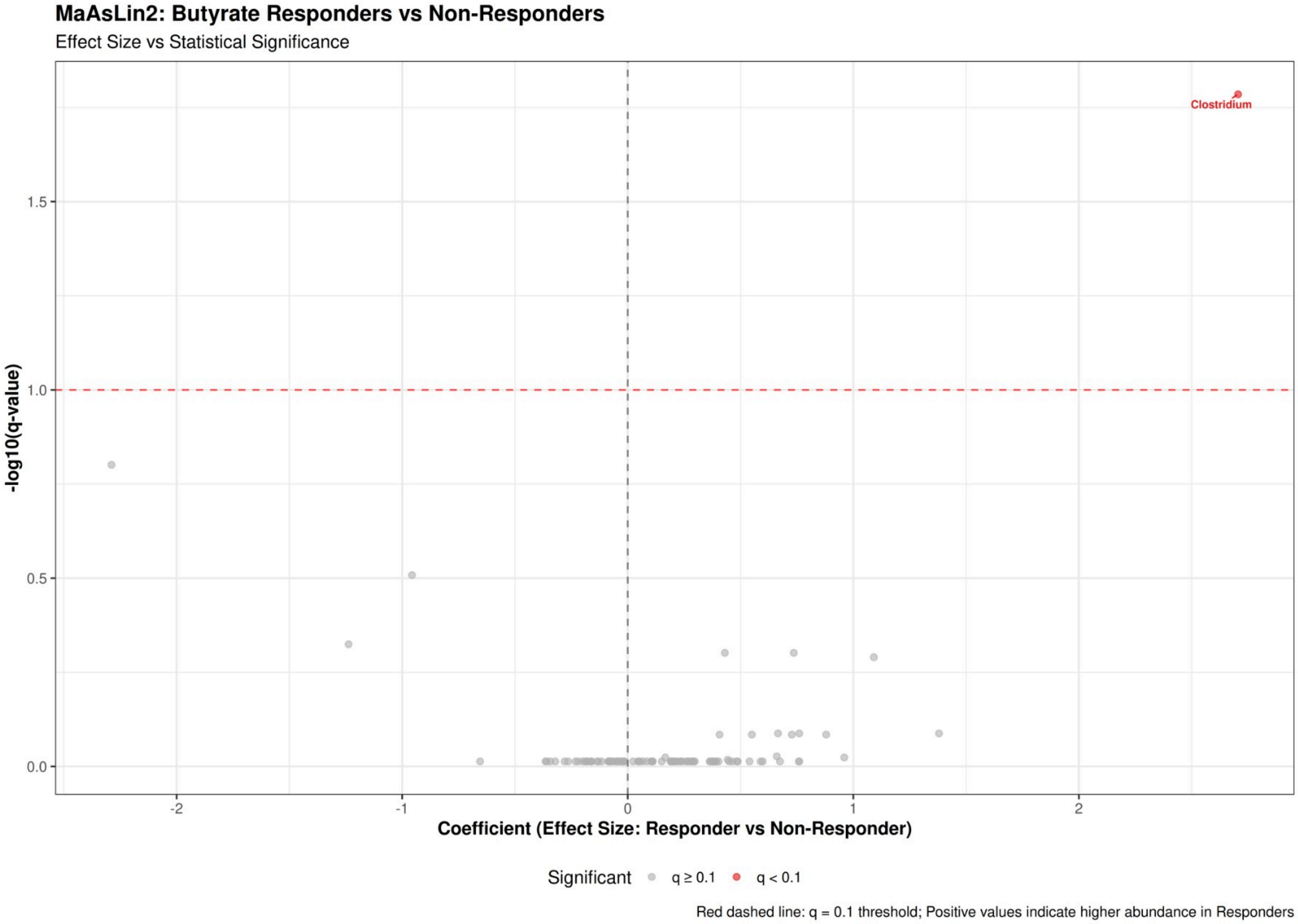
To identify specific taxa that differ between butyrate responders and non-responders, we performed differential abundance analysis using ANCOM-BC2 and MaAsLin2.

```
## ANCOM-BC2 butyrate responder: 0 significant taxa (q < 0.1)
```

```
## MaAsLin2 butyrate responder: 1 significant associations (q < 0.1)
```

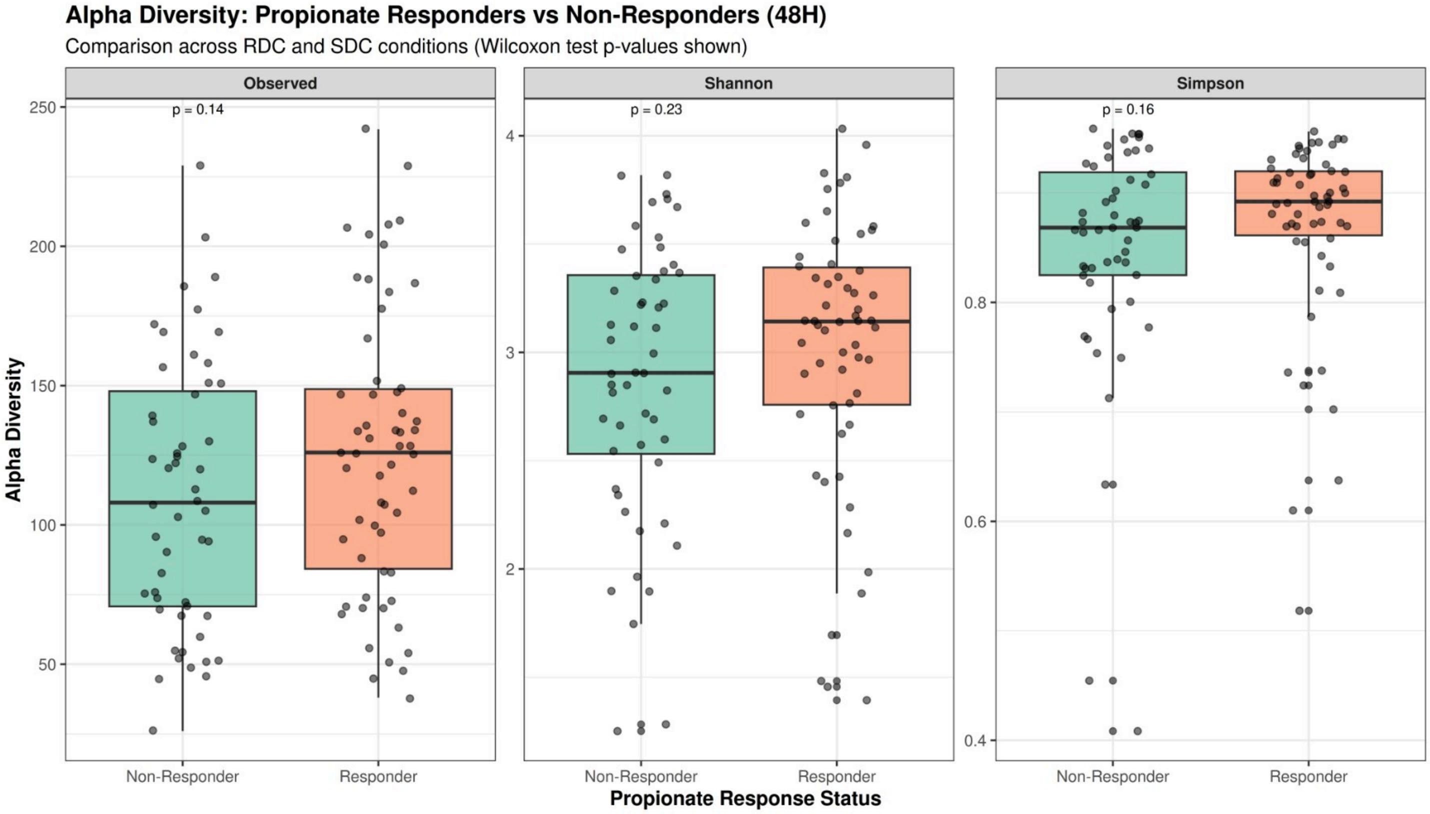
7.6.2 ANCOM-BC2 and MaAsLin2 Visualizations: Butyrate Responders vs Non-Responders



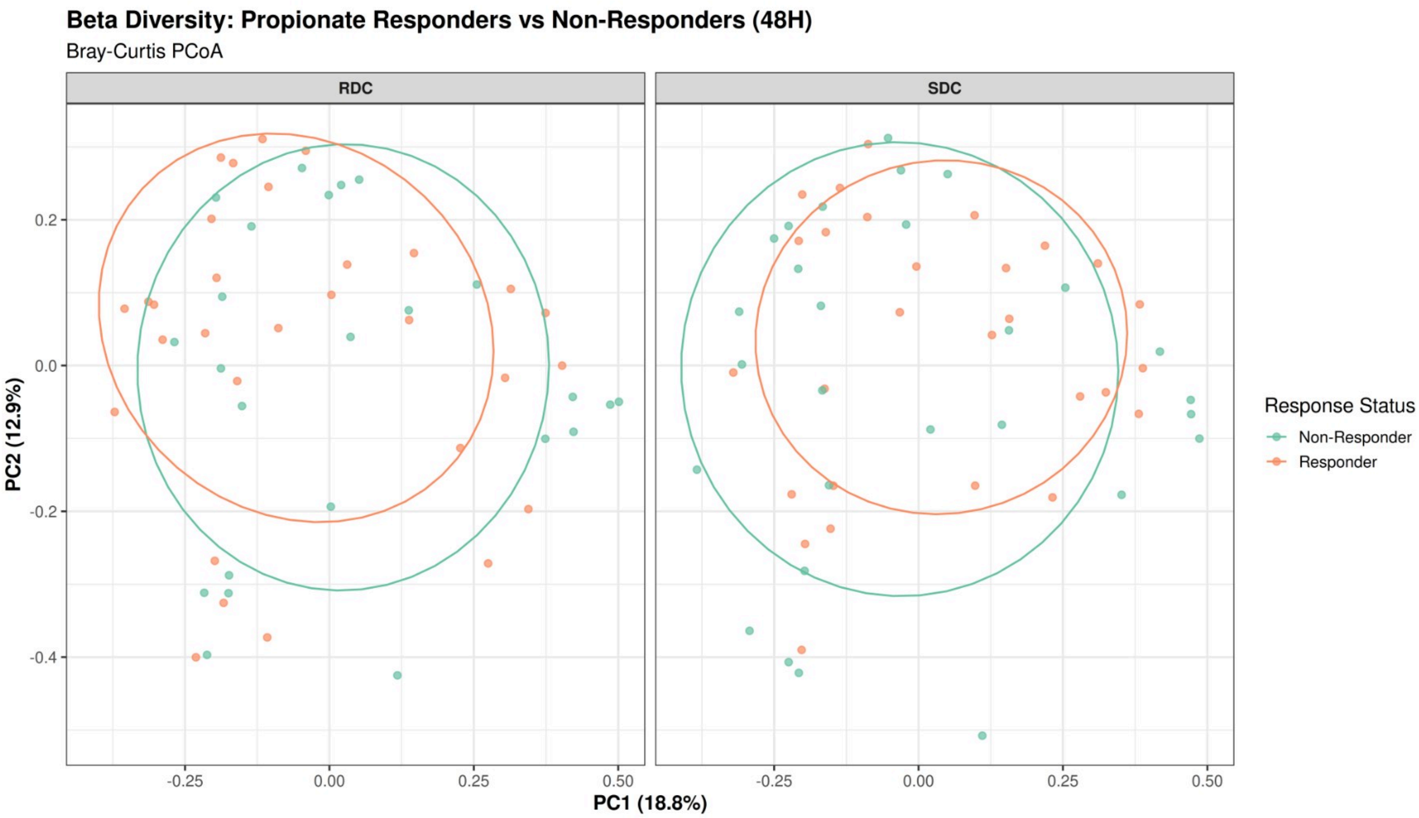


7.7 Microbial Composition Differences: Propionate Responders vs Non-Responders

Following the same responder classification approach described in Section 7.6, we compare the microbial composition between propionate responders and non-responders at the 48H timepoint. Propionate responder status was determined using a median split of the delta change in propionate concentration (48H - 0H) for each carbohydrate condition separately. This analysis examines whether individuals with greater propionate production responses have distinct microbial communities compared to those with smaller responses.




```
##
## Wilcoxon Test Results: Propionate Responders vs Non-Responders (Alpha Diversity)
## ================================================================================
## # A tibble: 3 × 6
##   Metric   Group1        Group2    `p value` `p adjusted` Significance
##   <chr>    <chr>         <chr>         <dbl>        <dbl> <chr>
## 1 Observed Non-Responder Responder     0.141        0.225 ns
## 2 Shannon  Non-Responder Responder     0.225        0.225 ns
## 3 Simpson  Non-Responder Responder     0.155        0.225 ns
```




```
##
## PERMANOVA: Propionate Responder Status (48H):
## Permutation test for adonis under reduced model
## Blocks:  strata
## Permutation: free
## Number of permutations: 999
##
## vegan::adonis2(formula = dist_bray_prop_resp ~ responder_propionate * carbohydrate, da
ta = meta_prop_resp, permutations = 999, strata = meta_prop_resp$subject_id)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      3    1.695 0.04939 1.8359  0.041 *
## Residual 106   32.614 0.95061
## Total    109   34.308 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

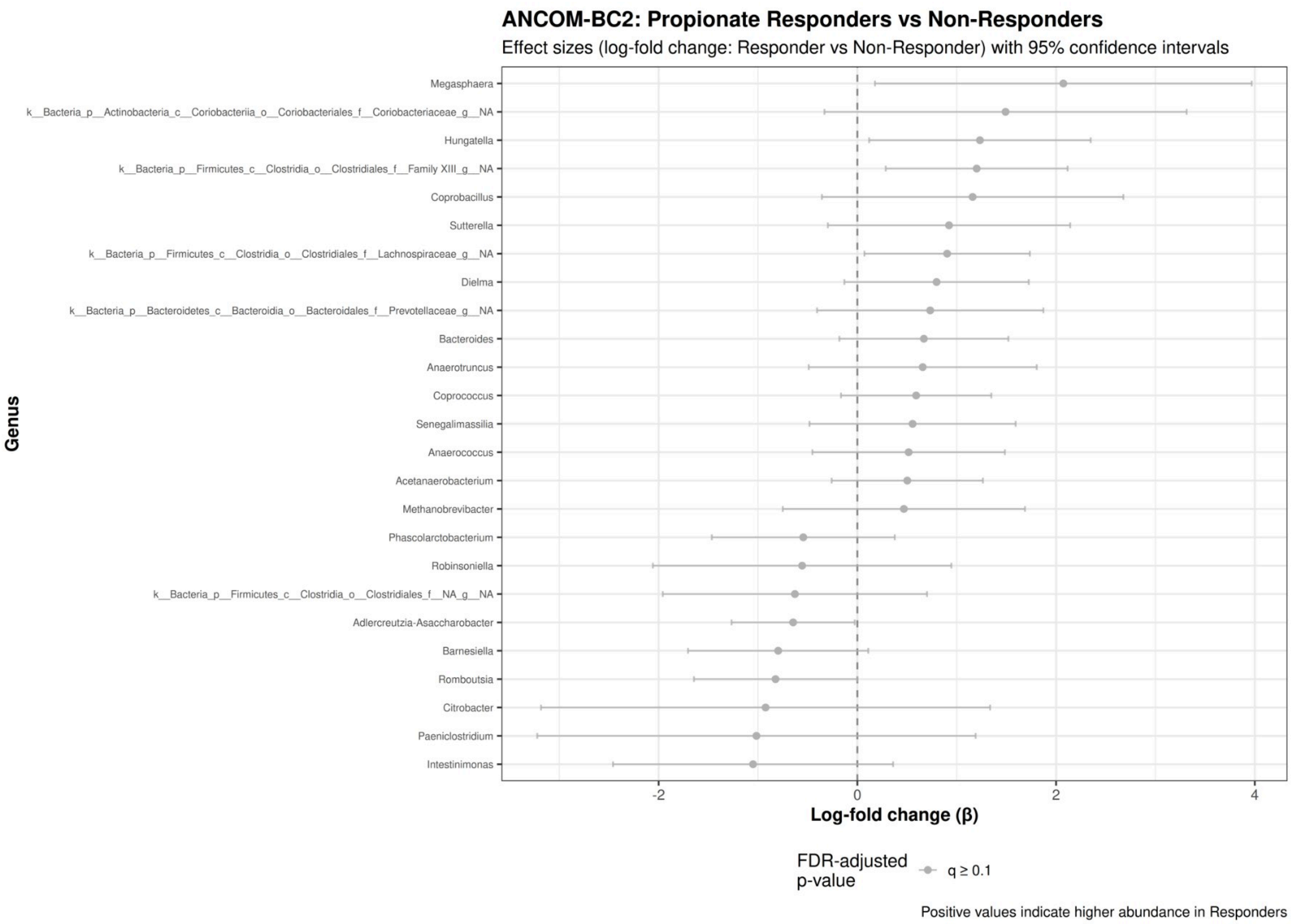
7.7.1 Differential Abundance Analysis: Propionate Responders vs Non-Responders

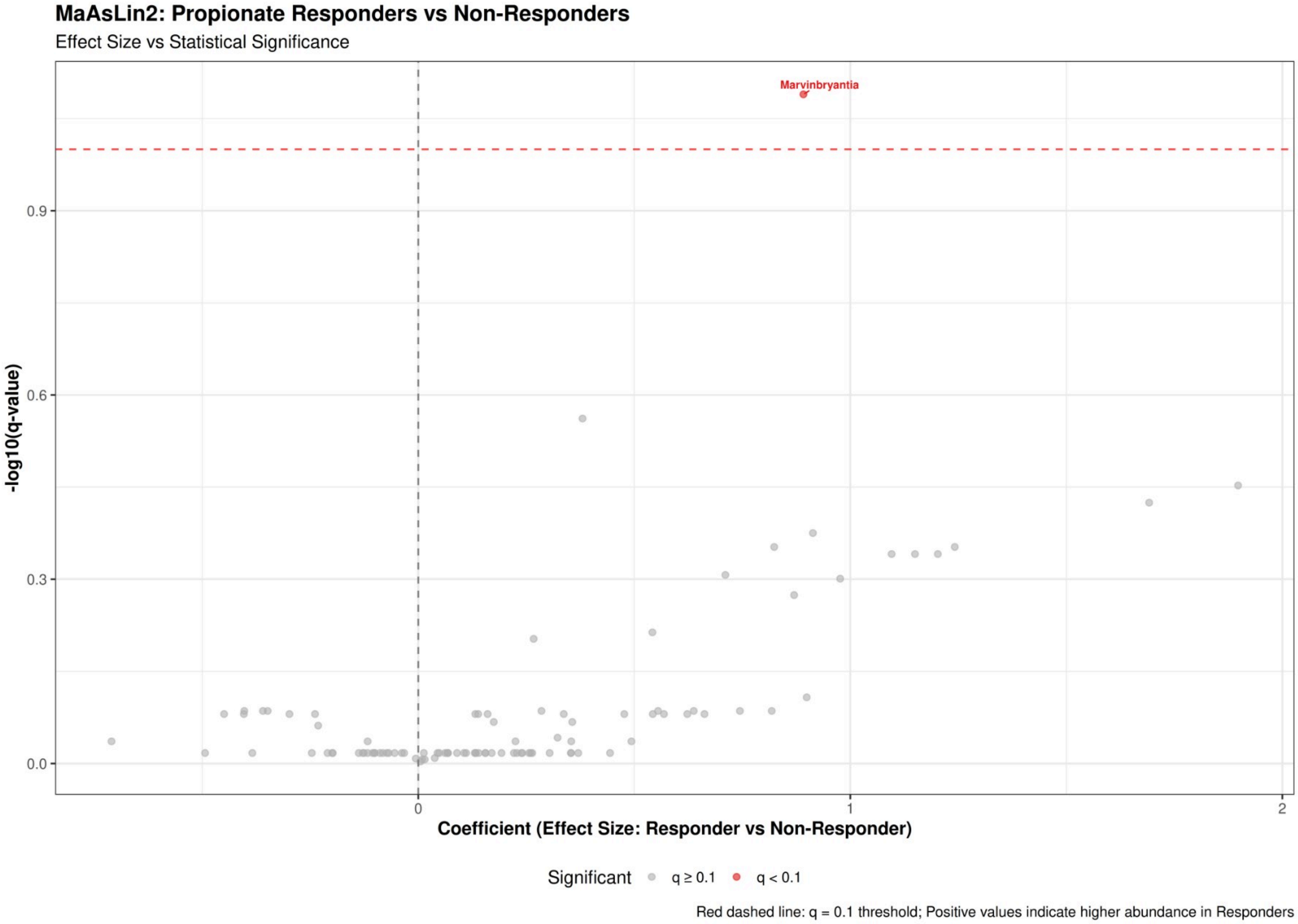
To identify specific taxa that differ between propionate responders and non-responders, we performed differential abundance analysis using ANCOM-BC2 and MaAsLin2.

```
## ANCOM-BC2 propionate responder: 0 significant taxa (q < 0.1)
```

```
## MaAsLin2 propionate responder: 1 significant associations (q < 0.1)
```

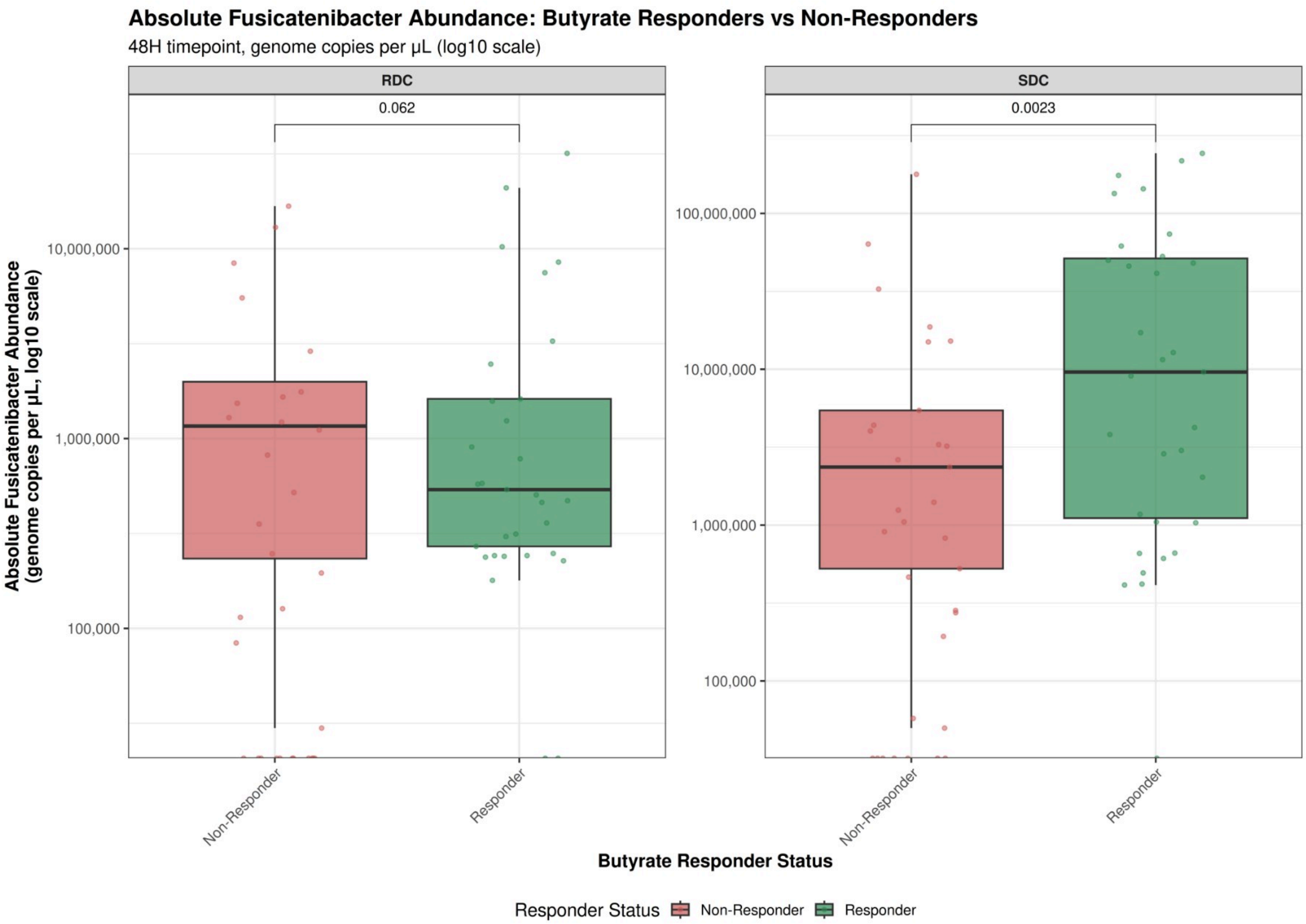
7.7.2 ANCOM-BC2 and MaAsLin2 Visualizations: Propionate Responders vs Non-Responders



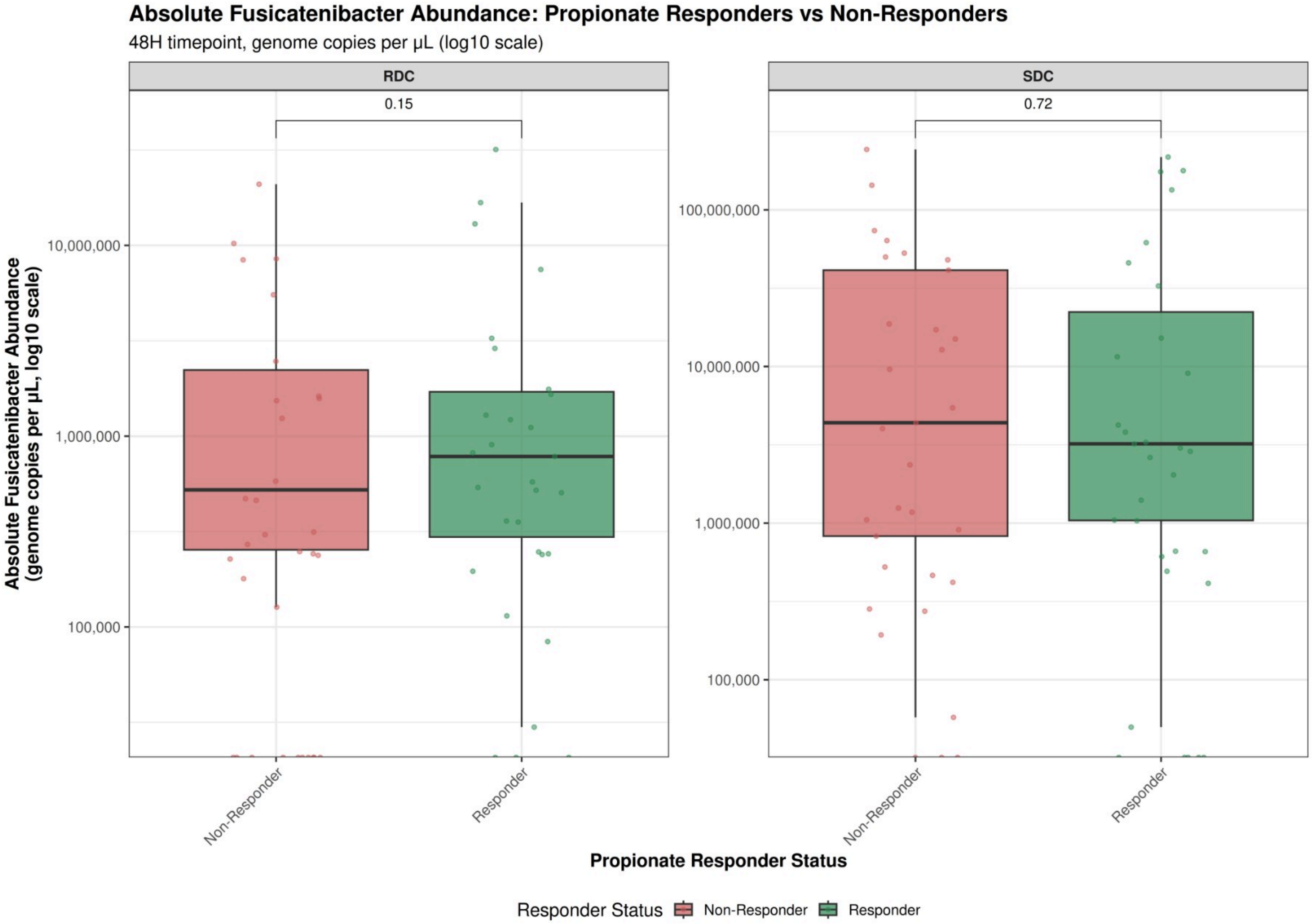


7.8 Absolute Fusicatenibacter Abundance: Responders vs Non-Responders

To determine whether absolute *Fusicatenibacter* abundance differs between SCFA responders and non-responders, we compared absolute abundance levels (genome copies per µL) at the 48H timepoint. This analysis uses qPCR-derived total bacterial load data integrated with relative abundance measurements to calculate absolute *Fusicatenibacter* abundance.



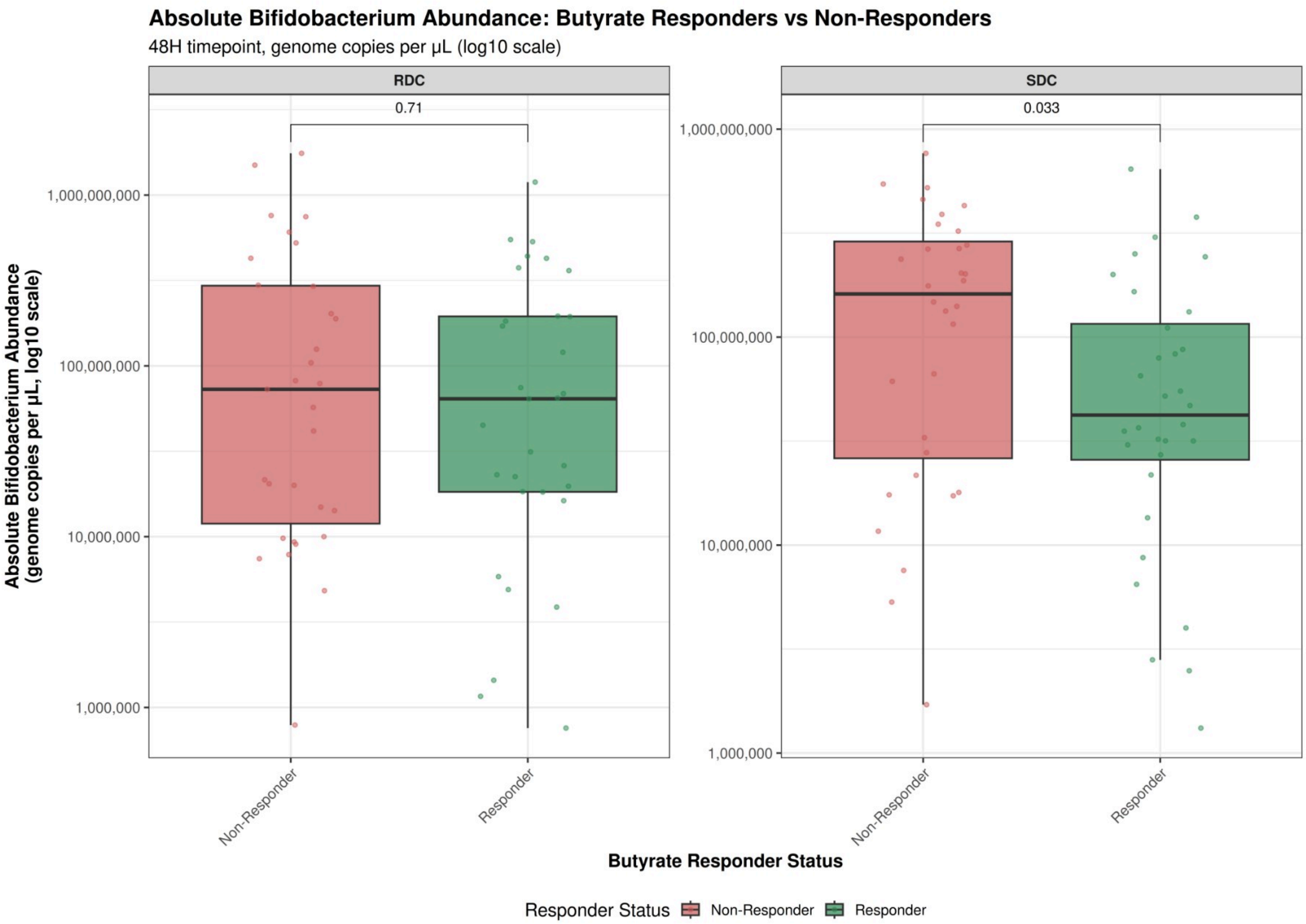

```
##
## ### Butyrate Responder Analysis:
## # A tibble: 4 × 6
##   responder_butyrate carbohydrate median_abs  mean_abs    se_abs     n
##   <chr>              <chr>             <dbl>     <dbl>     <dbl> <int>
## 1 Non-Responder      RDC             195794.  1858093.   707443.    31
## 2 Non-Responder      SDC             977671. 11121706.  5837706.    32
## 3 Responder          RDC             505336.  3123972.  1231200.    31
## 4 Responder          SDC            9329027. 42798736. 11817629.    32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                        group1        group2        n1    n2 statistic       p
## * <chr>                      <chr>         <chr>      <int> <int>     <dbl>   <dbl>
## 1 fusicatenibacter_abs_abund Non-Responder Responder     63    63      1323 0.00123
```



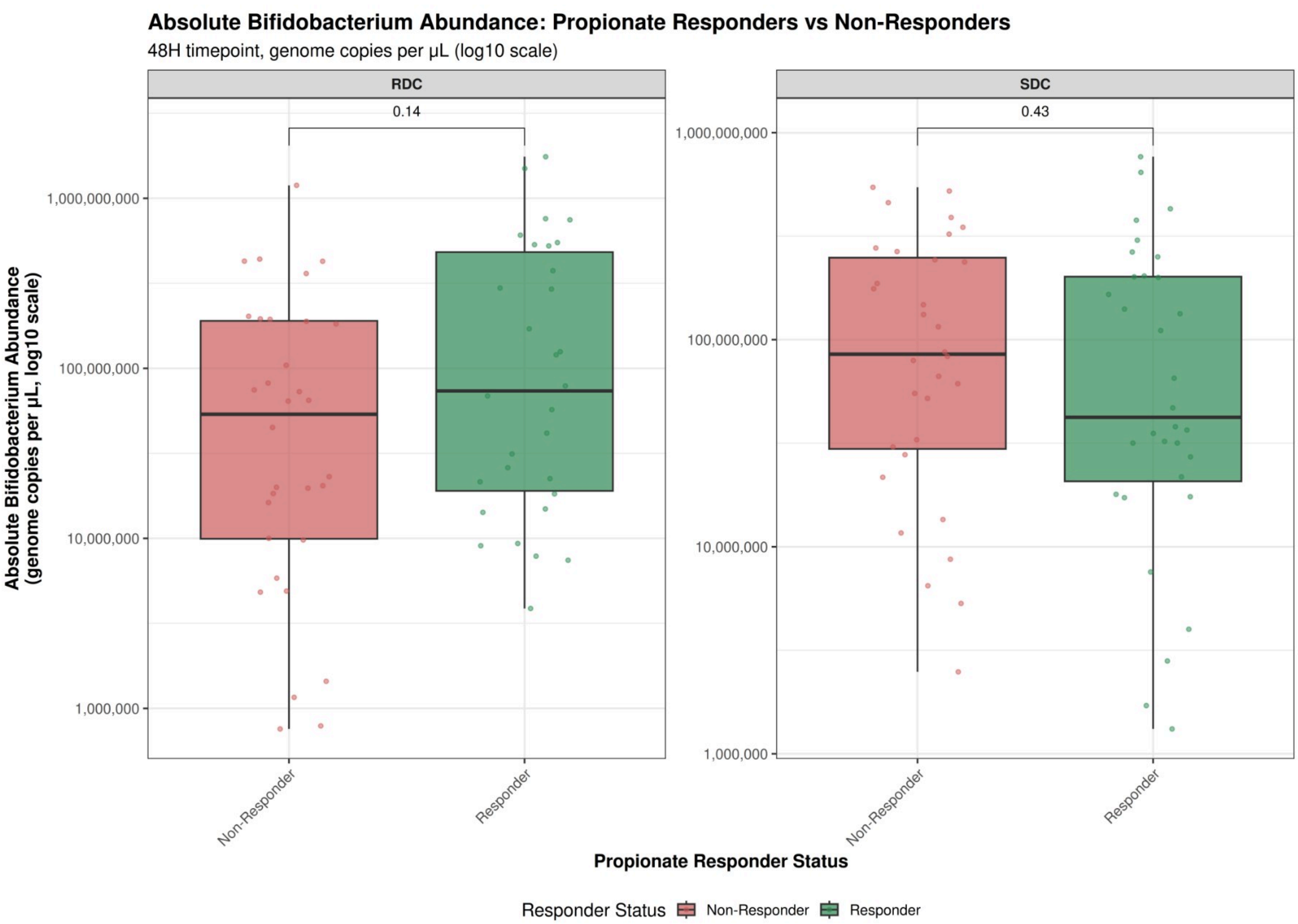

```
##
## ### Propionate Responder Analysis:
## # A tibble: 4 × 6
##   responder_propionate carbohydrate median_abs  mean_abs    se_abs     n
##   <chr>                <chr>             <dbl>     <dbl>     <dbl> <int>
## 1 Non-Responder        RDC             259365.  2053187.   776525.    32
## 2 Non-Responder        SDC            3185150. 25402241.  8924278.    32
## 3 Responder            RDC             556837.  2958068.  1217298.    30
## 4 Responder            SDC            2746683. 28518201. 10493819.    32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                        group1        group2       n1    n2 statistic     p
## * <chr>                      <chr>         <chr>     <int> <int>     <dbl> <dbl>
## 1 fusicatenibacter_abs_abund Non-Responder Responder    64    62      1853 0.523
```

7.9 Absolute Bifidobacterium Abundance: Responders vs Non-Responders

To determine whether absolute *Bifidobacterium* abundance differs between SCFA responders and non-responders, we compared absolute abundance levels (genome copies per μL) at the 48H timepoint. This analysis uses qPCR-derived total bacterial load data integrated with relative abundance measurements to calculate absolute *Bifidobacterium* abundance.



```
##
## ### Butyrate Responder Analysis:
## # A tibble: 4 × 6
##   responder_butyrate carbohydrate median_abs   mean_abs    se_abs     n
##   <chr>              <chr>             <dbl>      <dbl>     <dbl> <int>
## 1 Non-Responder      RDC            72946996. 258322171. 77015069.    31
## 2 Non-Responder      SDC           161960340. 200714144. 33989331.    32
## 3 Responder          RDC            64155493. 169288167. 45645934.    31
## 4 Responder          SDC            42425998. 100571916. 24285853.    32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                    group1        group2       n1    n2 statistic     p
## * <chr>                  <chr>         <chr>     <int> <int>     <dbl> <dbl>
## 1 bifidobacterium_abs_abund Non-Responder Responder    63    63      2319 0.103
```



```
##
## ### Propionate Responder Analysis:
## # A tibble: 4 × 6
##   responder_propionate carbohydrate median_abs   mean_abs    se_abs     n
##   <chr>                <chr>             <dbl>      <dbl>     <dbl> <int>
## 1 Non-Responder        RDC           54543035. 139742370. 41478245.    32
## 2 Non-Responder        SDC           85154019. 156785330. 28269346.    32
## 3 Responder            RDC           73777515. 292805487. 79562440.    30
## 4 Responder            SDC           42425998. 144500730. 33245101.    32
##
## Wilcoxon Test Results:
## # A tibble: 1 × 7
##   .y.                      group1        group2       n1    n2 statistic     p
## * <chr>                    <chr>         <chr>     <int> <int>     <dbl> <dbl>
## 1 bifidobacterium_abs_abund Non-Responder Responder    64    62      1855 0.531
```

8. Summary

This integrated analysis brings together SCFA metabolomics and 16S rRNA gene microbiome data to provide a comprehensive view of how carbohydrate interventions affect microbial metabolism and community structure in adolescents with and without obesity. Below we summarize the key findings from each major analysis component.

8.1 SCFA Metabolomics Findings

Carbohydrate Type Effects: - Both rapid digestible carbohydrate (RDC) and slow digestible carbohydrate (SDC) significantly increased SCFA production (butyrate, propionate, acetate) over 48 hours compared to no-carbohydrate controls. - Delta changes (48H - 0H) showed significant differences between carbohydrate conditions across multiple SCFAs, with both RDC and SDC stimulating substantial SCFA production. - The magnitude and timing of SCFA responses varied by carbohydrate type, as evidenced by significant carbohydrate × timepoint interactions in mixed-effects models.

Group Differences: - Limited significant differences were observed between Control (Healthy) and Case (Obese) groups in SCFA production capacity, suggesting that obesity status alone does not dramatically alter the fundamental ability of gut microbiota to ferment these carbohydrate substrates. - Mixed-effects models showed minimal significant main effects of group or major interactions involving group, indicating similar fermentation capacity between groups.

Interpersonal Variation: - Pronounced subject-to-subject variability in SCFA production was observed, with some individuals consistently showing high or low SCFA responses across carbohydrate conditions. - This variability was evident even after accounting for group and carbohydrate type effects, highlighting the need for personalized dietary approaches.

8.2 Microbiome Composition Findings

Alpha Diversity: - Alpha diversity metrics (Shannon, Simpson, Observed richness) showed significant differences between Healthy and Obese groups at the 48H timepoint, particularly in the no-carbohydrate control condition. - At baseline (0H), no significant differences in alpha diversity were observed between groups across carbohydrate conditions. - By 48H, Healthy individuals showed higher alpha diversity than Obese individuals in the no-carbohydrate condition, suggesting that carbohydrate challenge may reveal group differences in microbial diversity.

Beta Diversity and Community Structure: - PERMANOVA analyses demonstrated significant effects of group, carbohydrate type, and timepoint on microbial community composition (Bray-Curtis distance). - Community structure differences were most pronounced at 48H, with significant effects of both group and carbohydrate type on beta diversity. - The interaction between group and carbohydrate type was significant, indicating that the response to different carbohydrates varies by obesity status.

Differential Abundance: - ANCOM-BC2 and MaAsLin2 analyses identified multiple taxa that differed significantly between carbohydrate conditions, groups, and timepoints. - At 48H, significant differences were observed between RDC and SDC conditions, with distinct taxa associated with each carbohydrate type. - These findings suggest that carbohydrate structure and digestibility shape microbial community composition in predictable ways.

Absolute Abundance: - Integration of qPCR-derived total bacterial load data with relative abundance measurements enabled calculation of absolute abundances (genome copies per μL) for all taxa. - Absolute abundance analyses revealed patterns that complement relative abundance findings, distinguishing between true changes in microbial load versus compositional shifts. - Key taxa of interest, including *Fusicatenibacter* and *Bifidobacterium*, were examined using absolute abundance metrics to understand their true abundance changes in response to carbohydrate interventions.

8.3 Taxon–SCFA Relationships

Butyrate Associations (48H timepoint): - ANCOM-BC2 analysis identified multiple taxa associated with butyrate concentrations at 48H. - MaAsLin2 identified 11 significant associations between taxa and butyrate concentrations at 48H ($q < 0.1$). - These analyses focused on the 48H timepoint to capture associations at peak SCFA production following carbohydrate intervention.

Propionate Associations (48H timepoint): - ANCOM-BC2 analysis identified multiple taxa associated with propionate concentrations at 48H. - MaAsLin2 identified 11 significant associations between taxa and propionate concentrations at 48H ($q < 0.1$). - These analyses focused on the 48H timepoint to capture associations at peak SCFA production following carbohydrate intervention.

Fusicatenibacter–SCFA Correlations: - *Fusicatenibacter* relative abundance showed a significant correlation with butyrate concentration at 48H in the SDC condition ($R^2 = 0.037$, $p = 0.1569$). - *Fusicatenibacter* relative abundance also correlated with propionate concentration at 48H in the SDC condition ($R^2 = 0.004$, $p = 0.6546$). - These correlations were specific to the 48H timepoint and SDC carbohydrate condition, suggesting that *Fusicatenibacter* may play a role in SCFA production in response to slow-digestible carbohydrates.

8.4 Responder vs Non-Responder Analysis

Responder Classification: - Responder status was defined separately for butyrate and propionate based on median delta SCFA changes (48H - 0H) within each carbohydrate condition. - This data-driven approach identified individuals who showed substantial SCFA responses relative to their peers in the same condition.

Overlap in Responder Status: - Of 48 subjects with complete data, 12 (25.0%) were responders for both butyrate and propionate. - 12 (25.0%) were non-responders for both SCFAs. - This partial overlap suggests that butyrate and propionate responses are partially independent, with some individuals showing selective responses to one SCFA but not the other.

Microbial Composition Differences: - Alpha diversity comparisons between responder and non-responder groups at 48H revealed significant differences for both butyrate and propionate responders. - Differential abundance analyses (ANCOM-BC2 and MaAsLin2) identified distinct taxa associated with responder vs non-responder status for both butyrate and propionate. - These findings indicate that microbial community composition is predictive of SCFA response capacity, with specific taxa enriched in responder groups.

Absolute Abundance of Key Taxa in Responders: - Absolute abundance analyses (using qPCR-derived total bacterial load) revealed differences in *Fusicatenibacter* and *Bifidobacterium* abundance between responders and non-responders at 48H. - These analyses provide insights into whether responder status is associated with true changes in microbial load (absolute abundance) versus compositional shifts (relative abundance). - The comparison of absolute abundances helps distinguish between taxa that are truly more abundant in responders versus taxa that appear enriched due to compositional effects.

8.5 Integrated Conclusions

Key Insights: 1. **Carbohydrate structure matters:** Both RDC and SDC stimulate SCFA production, but the specific taxa and metabolic pathways involved differ, as evidenced by distinct differential abundance patterns and community structure changes.

2. **Obesity status has limited impact on fermentation capacity:** While some differences in baseline microbial diversity were observed, the fundamental ability to ferment carbohydrates and produce SCFAs appears similar between Healthy and Obese groups, suggesting that obesity-related microbiome changes may not directly impair carbohydrate fermentation.

3. **Individual variation is substantial:** The pronounced interpersonal variation in SCFA responses, combined with distinct microbial signatures in responder vs non-responder groups, highlights the importance of personalized nutrition approaches.

4. **Taxon–SCFA relationships are context-dependent:** Associations between specific taxa (e.g., *Fusicatenibacter*) and SCFA concentrations are strongest at specific timepoints (48H) and carbohydrate conditions (SDC), emphasizing the importance of considering temporal and dietary context in microbiome–metabolite relationships.

5. **Responder phenotypes have distinct microbial signatures:** The identification of microbial taxa associated with responder status suggests that microbiome composition may be predictive of individual SCFA response capacity, opening possibilities for microbiome-based precision nutrition strategies.

Clinical and Translational Implications: - The identification of responder vs non-responder groups, combined with distinct microbial signatures, suggests that microbiome profiling could inform personalized dietary recommendations. - The finding that *Fusicatenibacter* correlates with SCFA production in the SDC condition suggests that this taxon may be a key player in slow-digestible carbohydrate metabolism. - The partial independence of butyrate and propionate responder status indicates that different microbial pathways may drive production of these SCFAs, with implications for targeted dietary interventions. - Absolute abundance measurements provide critical context for understanding whether responder phenotypes reflect true increases in beneficial taxa (e.g., *Fusicatenibacter*, *Bifidobacterium*) or compositional shifts, informing strategies for microbiome modulation through dietary interventions.

Future Directions: - Functional metagenomics and metabolomics could identify the specific genes and pathways that drive SCFA production in responder groups. - Longitudinal studies could determine whether responder status is stable over time and whether dietary interventions can shift individuals from non-responder to responder status. - Clinical trials could test whether microbiome-based stratification improves the efficacy of dietary interventions for obesity management.